

USA Health Insurance Costs by Income (Quintile) Level, 2000-2023:

An Evaluation of Influential Factors

By:

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A Capstone Project Submitted to the Faculty of Utica University

January 2025

As required for fulfillment of Degree of Master of Science in Data

Science

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Abstract

Health insurance began in the United States (US) nearly a century ago, in the 1920s and 1930s (Smith, Katherine, 2023). Health insurance coverage and costs have been debated in the last few decades in the public arena and is an ongoing concern for many US citizens. (Political Science, 2024). Beginning in the early 1990s with the Clinton Administration’s proposal of the “Health Security Act” (CHA, 1993) to the passage and implementation of the Affordable Care Act(ACA) in 2010 (AANP, 2018), health insurance coverage and costs have been a focus of much heated public debate. This research project will evaluate factors influencing health insurance costs in the United States across different income groups during the 2000-2023 period. Health insurance cost influences include temporal factors such as government policy (specifically, the passage and implementation of the ACA) and global events (such as COVID-19), demographic, and regional factors. Significant demographic influences- such as age and location- were identified using regression and advanced machine learning analyses on data from the U.S. Bureau of Labor Statistics and the U.S. Census Bureau. Findings reveal that health insurance costs as a percentage of household expenses rose sharply for lower-income quintiles (Quintiles 1 and 2), with age ranges (65–74 and 35–44) and regional location (Metro In/Out) as key predictors. The ACA and COVID-19 pandemic also influenced health insurance costs, particularly among lower-income groups. By translating complex statistical findings into visualizations and actionable insights, this paper provides valuable knowledge for policymakers, insurers, news organizations, and the public, exemplifying key goals of data science: extracting insights, solving complex societal problems, predicting trends, and enhancing decision-making. (Sharma, 2024).

Keywords

Data Science, Dr. Jing-Mao Ho, quintiles, health insurance costs, temporal, multicollinearity, ridge regression.

Acknowledgments

For nearly three decades, my work experience has included many areas in data science, including health insurance, state and federal government agencies, financial institutions, and medical device/pharmaceutical companies. My roles and duties include statistical research; data extract, transfer, load (ETL); data reporting; health insurance claims processing systems implementation, administration, and reporting; and statistical software (SAS) development and administration. Given this broad background in many diverse fields of data science, I am in a unique position to analyze the costs of health insurance by income group and offer some key insights based on the knowledge I have gained from being allowed the opportunity to obtain my master's in data science from Utica University. Thank you to Dr. Michael McCarthy for working with me on admission, coursework, and this capstone project direction. Also, thank you to Dr. Jing Ho- for working with me on coursework and this capstone project. In remembrance of, and also special thanks to my high school math teacher, Mr. Terrance Balk, who understood and encouraged me to develop my math skills, despite my shortcomings.

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Introduction

Health insurance costs significantly impact household financial stability, accessibility to healthcare services, and national economic health. In recent decades, escalating healthcare expenditures have become central to American economic and social policy debates. Policies such as the Affordable Care Act (ACA), enacted in 2010 and fully implemented by 2015, aimed at addressing these rising costs and increasing coverage (111th Congress, 2009-2010).

Additionally, global events such as the COVID-19 pandemic have introduced unprecedented pressures on healthcare systems, further complicating the understanding and managing health insurance costs. Thus, analyzing factors contributing to the rising cost of health insurance, particularly across different demographic and income groups, is essential for informed policy-making and economic planning.

Numerous statistical and other formal studies have been conducted measuring the effectiveness of the ACA with its intent to increase the number of individuals with health insurance coverage in the United States. Many studies have validated that the number of individuals with coverage in total and as a percentage of the total population has increased substantially. Several additional studies have been undertaken verifying or researching the ACA's impact on healthcare coverage of various demographic groups, specific age groups, specific medical conditions, and employment types. While studies confirm increased coverage, gaps remain in understanding cost impacts across income quintiles, especially alongside COVID-19 and regional/demographic factors.

Problem Statement

This research investigates the impacts of ACA policy implementation, demographic and regional characteristics, and the COVID-19 pandemic on health insurance costs—as a percentage of household expenditures and, in absolute terms, adjusted to 2023 USD—for various income groups in the United States between 2000 and 2023. Identifying and quantifying these influences will clarify the dynamics driving health insurance expenditures, informing policymakers, economists, and healthcare stakeholders.

Scope

The Scope of this project includes the evaluation of health insurance costs as a percentage of total household expenses by income group (quintile, defined as follows: Low Income {lowest 20% of all income earners}, Lower Middle Income {next higher 20% of all income earners}, Middle Income { middle 20% of all income earners}, Upper Middle Income {second from highest 20% of all income earners}, and Upper Income {top 20% of all income earners} over a nearly forty year period (1984-2023). Total health insurance costs by year from 2000 to 2023 by income quintile were also obtained and adjusted for 2023 United States Dollars (USD\$) to be used as an additional dependent variable to be analyzed. The regional variables include Metro Location and Region. The demographic variables include Race, Household, Age, Employed Number of Jobs, and Work Type (full versus part-time). A more detailed list and description of all variables used in the project are listed in "SECTION 1: Data" below.

Research Methods

The approach employed for the research conducted to answer this research question closely followed the Cross-Industry Standard Process for Data Mining (CRISP-DM). (Shearer, 2000). Following this methodology, the report is organized into the following four sections, listed below. This report coincides with the Data-Information-Knowledge-Wisdom (D-I-K-W) framework approach used in many data science studies (Berengueres, Sandell, 2020)

Section 1: Data - Define and Wrangle (Data)

Supply detailed definitions of data sets used in the report.

Account for missing data values. Join data sets.

Section 2: Data – Describe (Information)

Provide summary statistics and correlations of the data.

Produce ANOVA tables and perform relevant hypotheses testing, if applicable.

Section 3: Data – Model (Knowledge)

Simple Linear Regression, Multiple Linear Regression, and

Machine Learning Algorithms (Random-Forest and Ridge Regression)

- For evaluation of Regional and Demographic variables

ANCOVA – for evaluation of temporal independent variables

Section 4: Data – Findings (Wisdom)

Note: For this project, all analysis, summary statistics, correlations, regression testing, machine learning, tables, and graphs were produced using R software version 4.4.0 (2024-04-24), nicknamed "Puppy Cup." MS Excel was used to assist in formatting the tables.

ChatGPT was used to find the regional and demographic data from the US Census

Bureau. ChatGPT was also used to assist with R coding and analysis. (OpenAI, 2025),

Ethics/ Transparency/ Bias/ Social Responsibility

The ACA has impacted nearly all US Citizens. This article may be used in academic environments, by interested individuals, or in congressional reviews by politicians interested in learning how the legislation has impacted different income groups. With this in mind, the author of this study took careful consideration to ensure that all data used is from credible sources and that the statistical methods of review and validation are carefully peer-reviewed for authentication, underscoring the commitment to ethical research practices. The author used the methods the National Academies of Sciences, Engineering, and Medicine utilized in the approach and application of methodology throughout this project. (NASEM, 2019)

Ethics

From an ethical perspective for this Health Insurance Cost Study, the author painstakingly took steps to ensure unbiased data sources were used and the methodologies used were undertaken with responsibility and integrity. Special consideration was given to data source respect and the broader social impact of this research.

Transparency

The author's two primary data sources for this research were the Bureau of Labor Statistics (Health Insurance and All Household Cost Data, by US Income Quintiles) and the US Census Bureau (Demographic and Regional variables, by US Income Quintiles). The BLS Consumer Price Index (CPI) was also used to compute the Health Insurance Cost variable values, adjusted for 2023. All data used in this paper were obtained from reputable national surveys conducted by the appropriate US Federal Government Agency, using scientific data collection and statistical methods to ensure that the data used is representative of the US population as a whole. Government agency data ensured consistent methodology for

standardized data collection and reporting methods were utilized. BLS consumer expenditure data and the Census Bureau's HINC-05 demographic tables are readily available to the general public. The period of this study, 2000-2023, also considers the impact of significant policy changes (ACA implementation) and economic incidences (COVID-19 Pandemic) on health insurance costs.

Bias

Among the strengths of this Health Insurance Cost Study include utilizing publicly available data from reputable sources to ensure credibility. Transparency is addressed by considering how variables were derived, included, excluded, and why. The use of multiple modeling approaches such as simple linear regression for year and quintile views, analysis of the demographic and regional data utilizing several machine learning techniques (Random Forest, Ridge Regression, and a combination of these two methods), and the ANCOVA process to gain insight into the temporal categorical variables ACA and COVID-19 reduced the reliance on a single methodology for analysis.

Social Responsibility

By utilizing publicly available data from trusted government sources (BLS and USCB) that are representative of the entire US Population, a socially responsible approach to this study has been transparently communicated throughout this document. A detailed discussion of how demographic variables were considered and calculated also underscores the commitment to social responsibility throughout. The explicit examination of quintile health insurance costs sheds light on the financial burden experienced by individuals in lower-income groups. The well-documented methodology of traditional regression techniques, including machine learning, emphasizes the aim of reducing bias and increasing the accuracy of the findings of this report.

Literature Review

Data Source Credibility, Combining Different Survey Datasets for Analysis:

The Bureau of Labor Statistics (BLS) is the leading federal agency for collecting, analyzing, and publishing economic data, leading to great credibility. The agency's consumer expenditure surveys focus on income, expenditures, and demographic characteristics of United States(US) consumers. The data is rigorously collected using statistically valid procedures to represent the US population. The BLS was the source for this study's health insurance and total household income data by income quintile and year. (BLS, 2025)

Using scientifically appropriate sampling and weighting techniques, the US Census Bureau is the official source of demographic, economic, and regional data. The tables used for this study's demographic and regional data were sourced from "HINC-05. Percent Distribution of Households by Characteristics". These datasets are explicitly designed to analyze household characteristics and income distributions. The US Census Bureau is another highly credible data source, often cited in peer-reviewed research and policy analysis. (USCB, 2025)

Combining datasets from the US Bureau of Labor Statistics and the US Census Bureau for statistical analysis is common, provided the datasets share standard variables for merging and integration. This study uses year and income quintile as the merge or join variables. Three important considerations for combining datasets from different survey sources include standard identifiers, data consistency, and methodological review and alignment. Best practices include data cleansing or wrangling, documentation, and applying appropriate statistical methods. Both sources are considered among the most reliable statistical agencies in the US, and combining their datasets meets the goals of established research practices and methodologies. (National Academies Press, 2017)

Combining Income Quintiles/Strata to Represent a Weighted Measure

Combining income strata (quintiles or five ranges representing twenty percent of the population) is common in government agencies such as the Bureau of Labor Statistics or the US Census Bureau. (BLS, October, 2023) This method was used to chart an overall view of health insurance costs as a percentage of all household expenses and as a raw dollar value.

Additionally, subsequent analyses such as summary statistics and correlations were produced where all income ranges were combined to give a "sense of direction" to what direction healthcare costs have been trending since 2000.

Predictive Methodology Utilizing Multicollinear Independent Variables

After ascertaining the appropriate demographic and regional independent variables for this research project, it was noted that all categories (seven categories utilized in this report) were interdependent or summed to 100% within each group of interest. Multicollinearity is defined by two or more independent variables in the regression model that are highly correlated. A key indicator is a high pairwise correlation between independent variables (for example, Pearson $r > 0.7$) (Sawtooth Software, 2024). For example, this project's seven categories used for analysis were metro location, region, race, household definition, age, employed (number of jobs), and work type (full-time, part-time, and no work).

Recommended best modeling methods for multicollinear independent variables include multiple regression with adjustments, logistic regression, and machine learning models. Some examples of adjustments for multiple regression include removing one of the variables (i.e., removing one of the four regions (Northeast, Southwest, Midwest, or West) from the analysis and reviewing analysis via ANOVA or correlation techniques (Pearson). (Penn State 2018) However, Pearson and ANOVA tables may only give estimates of relationships and

should not be solely relied upon for selecting variable relationships. Another test, the Spearman rank correlation, could be considered for data that is continuous and nonnormally distributed, ordinal, or with relevant outliers and can be used as a measure of a monotonic association. (Schober, Boer, Schwarte, May 2018) A monotonic relationship is a relationship in which either variables increase simultaneously or are related inversely (as the value of one variable increases, the other variable value decreases) (Lund, 2018).

Two recommended Machine Learning methods to overcome multicollinearity in statistical modeling include tree-based models such as Random-Forest(RF) and Ridge Regression. The tree-based methods overcome multicollinearity without relying on linear relationships. Ridge Regression regularizes coefficients in the model and helps to stabilize the model when the predictor variables are colinear. (Garg, Tai, 2013) A combination of the two methods was first employed, beginning with employing Random-Forest to narrow the list of available independent variables in the project. This method considers two measures for retaining or ranking variables regarding model relevance. The first is the percent increase in mean square error (%IncMSE) of a tree, and the next is the increase in node purity (IncNodePurity), which is noted as a decrease in node impurity. (Genuer, Poggi, Tuleau-Malot, 2010). Other or additional regularization techniques (such as Ridge Regression) are suggested to occur after variable selection to refine the model further. (Boulesteix, Janitza, Kruppa, & König, 2012). Random Forest ranks variables based on their importance relative to outcome predictions, and the %IncMSE metric is a precise measure of the variable portion of model predictability. RF works well for narrowing down many independent predictor variables, such as the phenomena of over thirty independent variables used for this project. Ridge then provides comprehensible

coefficients for each variable. Ridge Regression is unlike RF, which has more of a black box or unknown predictor value.

Use of Temporal Predictor Variables in Modeling

In this examination, two independent categorical variables, ACA and COVID1 (COVID-19), are defined by periods spanning several years and are, therefore, referred to as temporal variables. (Tang, Peng, Liu, Zhang, 2010) Because ACA and COVID-19 are defined by year, this also leads to the variables being defined as being multicollinear. Due to this multicollinearity, care must be taken when analyzing or including these variables in modeling and ANOVA analysis.

Performing ANOVA analysis may be useful if Year and Quintile are used (excluding demographic and regional variables due to multicollinearity within each variable type). Also, ANOVA could be performed with ACA and Quintile or ACA and COVID-19 (excluding year and demographic and regional variables) (Kutner, Nachtsheim, Neter, and Li; 2005) and the dependent health insurance percentage and amount variables.

ANCOVA combines the strengths of regression analysis and ANOVA by adjusting for covariates and effectively controls confounding variables (in this case, YEAR). (Philippas, 2014) ANCOVA analysis will be performed for the temporal variables used in this report, ACA and COVID1.

Factors Impacting Health Insurance Costs

Affordable Care Act (ACA)

In the initial phase of the ACA, the primary focus was on extending insurance coverage to more people. Two additional areas will need to be focused on in the future, including slowing

the increase in health costs and improving the physician's practice environment. (Cutler, 2015)

This second area, slowing the increase in health costs, is the focus of this document.

COVID-19

During COVID-19, fluctuating premium patterns were noted. For example, in 2021, insurers adjusted changes ranging from a 3.4% decrease to an 8.4% increase to accommodate the anticipated impact of COVID-19. (McDermott, Kurani, Ramirez, Shanosky, and Cox, 2020) Significant job losses were incurred during the pandemic, resulting in many Americans losing health care coverage offered by employers. To a limited extent, the decline was partially counter-balanced by a noted increase in enrollment in Medicaid and the Health Insurance Marketplace during the COVID period (McDermott, Cox, Rudowitz, Garfield, 2020).

Demographics

Age, marital status, education level, and the log of salary income were associated with higher family spending on health insurance premiums and medical expenses. Data from the 2014 Consumer Expenditures Survey was used in this research project by Choi and Blackburn (Choi, Blackburn; 2018). The authors of this article had a similar goal in mind (measuring the impact of demographics on health insurance costs) as the focus of this paper's analysis. The authors limited their study to one year. They did not break out the analysis by income quintile. However, in this paper's analysis, they used the "log of salary income" as an independent predictor variable rather than a "group by" variable.

Additional Healthcare Cost Studies

The report, "Consumption of Healthcare Services in the United States: The Impact of Health Insurance," conducted in 2023, notes that healthcare expenditures have increased due to several factors. These influences include differences in insurance plans, demographic factors

including age, race/ethnicity, and citizenship, and the specific health service required. Historical data, including the RAND Health Insurance Experiment, were reviewed to determine how insurance changes consumption choices and drives costs. Some key findings reported included rising costs due to moral hazards (overused when more insurance is available), such as health insurance use increases, causing health utilization increases. Age factors such as older Medicare patients consume more services, driving up costs. The findings of this Capstone project coincide with this study, as age cohorts were found to be correlated with increased health insurance costs. (Narita, 2023)

The article "The Complex Relationship between Cost and Quality in US Health Care" notes the extreme expense of healthcare costs in the US. Per capita spending on healthcare in the US is 1.5 to 3 times higher compared to other economically developed countries. The US leads the world in healthcare costs but ranks twenty-sixth for life expectancy. Researchers have attempted to identify wasteful spending that leads to higher healthcare costs and found that elements such as varying prices for identical services depending on insurer negotiations, administrative costs (billing, insurance overhead), and policies requiring coordination between private and public payors contribute substantially to increased costs. This research project took a different approach to determine health insurance cost increases based more on the administration of health care as a driving factor for increased costs as opposed to demographic or temporal variable considerations. This approach helps provide a more comprehensive understanding of factors influencing overall health insurance and healthcare costs. (Burke, Ryan, 2014) Another research project, "Projected costs of single-payer healthcare financing in the United States: A Systematic Review of Economic Analyses" also identified administrative costs, health care utilization, and drug pricing as influencers of health care cost increases. A key finding from this

report found that "increased utilization from universal coverage raises costs by 2.0%–19.3% (median 9.3%) due to newly insured patients accessing care." (Cai et al., 2020)

Machine Learning and Healthcare Cost Studies

Machine Learning approaches to understanding health insurance cost factors were also researched in additional articles. "Machine Learning-Based Regression Framework to Predict Health Insurance Premiums" focused on machine learning (ML) regression models to predict health insurance costs utilizing demographic variables such as age, children, smoking status, BMI, region, and sex. All independent variables were used in the predictive model prediction. A linear relation between age and healthcare charges was identified. A linear regression model with an R² of .75 versus an artificial neural network (ANN) utilizing gradient boosting with an R² value of .9272 was determined from the study. Age was also an important contributing factor identified in the Capstone focus for the machine learning approach. (Kaushik, Bhardwaj, Dwivedi, Singh, 2022).

In another machine learning data project, "Machine Learning for an Explainable Cost Prediction of Medical Insurance," three different models, including Extreme Gradient Boosting (XGBoost), Gradient Boosting Machine (GBM), and Random Forest (RF), were used to predict medical insurance costs. Key variables impacting costs were found to be age and region (equivalent to the findings of this Capstone) in addition to body mass index (BMI), smoking status, and number of children, similar to the variables in the previous machine learning report discussed above (Orji, Ukwandu, March 2024).

METHODOLOGY

SECTION 1: Data - Define, Wrangle (D-I-K-W: Data)

Two primary data sources were utilized for the analysis performed for this project. For this study, meticulously gathered data sets with health insurance expenses and total household expenses by income quintile group over the approximate twenty-three-year period were obtained. The data was obtained from the following primary two sources: the Federal Reserve Bank of St. Louis, FRED: Economic Data, Health Insurance by Quintiles of Income Before Taxes and the US Bureau of Labor Statistics (US BLS), Expenditures: Total Average Annual Expenditures by Quintiles of Income Before Taxes for the years 1984-2023.

Data sets with extensive information about the demographic components of the five income quintile groups were also obtained from the US Census Bureau (USCB) from 2000 through 2023. Seven key additional variables garnered from this site used in this analysis include Metro Location, Region, Race, Household, Age, Employed Number of Jobs, and Work Type (full vs. part-time). The report is entitled "Household Income (for the year of interest)," and the data utilized in this analysis is entitled "Percent Distribution of Households, by Selected Characteristics Within Income Quintes," and it is made available every year.

The cleansed datasets were joined (by year and income quintile) as needed. The methodology used to combine the datasets followed well-established industry practices (National Academies Press, 2017).

In addition to accessing the periods in which the ACA was implemented and took effect, the periods surrounding the COVID-19 pandemic were also included as a potential independent explanatory variable used to measure the event's impact on health insurance costs.

Additionally, the years defining the periods of interest for the ACA (ACA_pre: 2000-2009, ACA_Implement: 2010-2014, and ACA_Post: 2015-2023) were assigned in the data set (111th Congress, 2009-2010). The years for COVID-19 included CVD_Pre 2000-2019 and CVD_Pandemic 2020-2023 (SOURCE XYZ2). Specific data variables in this research project are defined below in Table 1 – Variable Descriptions. The R data frame name is dfACA01.

Table 1 - Variable Descriptions

VarType/Group	
Group Bv/Join X Variables:	
Full Variable Name Description	Dataset Trim Variable Name
Year	Year
Quintile (Income quintile by 5 groups)	Qnt1, Qnt2, Qnt3, Qnt4, Qnt5
Y Var Values:	
Full Variable Name Description	Dataset Trim Variable Name
Expense Variables in \$ Amounts	
Expense All	Exp USD ALL
Expense Health All	Exp USD Hlth All
Index Years to 2023 USD	Conv 2023 USD
Expense Health USD All	Exp USD Hlth ALL 2023 Adj
Expense USD Health All 2023 Adj	Exp USD Hlth Ins
** Expense USD Health Insurance 2023 Adj	Exp USD Hlth Ins 2023 Adj
Expense Variables as % of All Household Income	
**Expense Percent Health Insurance	Exp Pct Hlth Ins
Expense Percent Health All	Exp Pct Hlth All
** Indicates Variables Used for Dependent Y values in modeling	
Temporal/Categorical X Vars:	
Full Variable Name Description	Dataset Value
Xc1 Affordable Care Act (ACA) was divided into three periods	
Before ACA (2000-2009)	ACA Pre
Implementation ACA (2010-2014)	ACA Impl
Post ACA (2015-2023)	ACA Post
Xc2 COVID-19 (COVID1) was divided into three periods	
Before COVID-19 (2000-2019)	CVD Pre
Pandemic COVID-19 (2020-2023)	CVD Pndmc
Xc3 COVID-19 (COVID2) was divided into three periods	
Before COVID-19 (2000-2019)	CVD Pre
Pandemic COVID-19 (2020-2022)	CVD Pndmc
Post COVID-19 (2023)	CVD Post
X1-X7: Demographic/Regional Variables:	
Each Data Var Represents Percents per Category	
Full X Variable Name Description	Dataset Trim Variable Name
X1 - MetroLocation (Metro In - Out, Metro Centarl City In-Out)	M_IN, M_CC_IN, M_CC_OUT, M_OUT
X2 - Region (NorthEast, MidWest, South, West)	RE_NE, RE_MW, RE_S, RE_W
X3 - Race (White, Black, Asian, Hispanic)	RC_WH, RC_BL, RC_AS, RC_HS
X4 - HouseHold (Family, Married, Non Family)	HH_FAM, HH_MAR, HH_NFM
X5 - Age (Less than 65, Greater Then-Equal 65 LowestAgeValue And Up... example 15 and Up=15U)	AG_LT_65, AG_GE_65; AG_15U, AG_25U, AG_35U, AG_45U, AG_55U, AG_65U, AG_75U
X6 - Employ Earners (Eamers, 1-4 Earners, No Earners)	EMP_NONE, EMP_1, EMP_GT2, EMP_2, EMP_3, EMP_4
X7 - WorkType (Total, Full Time, Part Time, No Work)	WRK_TOT, WRK_FT, WRK_PT, WRK_NO
NOTES:	
X1 - NOTE_ML: M_IN + M_OUT ~ 100%; M_CC_IN + M_CC_OUT + M_OUT ~ 100%	
X2 - NOTE_RG: RE_NE + RE_MW + RE_S + RE_W ~ 100%	
X3 - NOTE_RC: RC_WH + RC_BL + RC_HS ~ 98% to 114% (adjusted to 100% in R-Code)	
X4 - NOTE_HH: HH_FAM + HH_NFM ~ 100%	
X5 - NOTE_AG1: AG_LT_65 + AG_GE_65	
X5 - NOTE_AG2: AG_15U + AG_25U + AG_35U + AG_45U + AG_55U + AG_65U + AG_75U ~ 100%	
X6 - NOTE_E1: EMP_NONE + EMP_1 + EMP_GT2 ~ 100%	
X6 - NOTE_E2: EMP_NONE + EMP_1 + EMP_2 + EMP_3 + EMP_4 ~ 100%	
X7 - NOTE_W1: WRK_TOT + WRK_NO ~ 100%	
X7 - NOTE_W2: WRK_FT + WRK_PT + WRK_NO ~ 100%	

Table 1 Variable Descriptions

Additional Data Notes:

Demographic percentages calculated were the same as the vertical percentages obtained from the sources, within ~99.9% compared with the vertical percentages. For Metro Year 2004, no data was available for the four categories X1-5 quintiles. The average of the previous four years (2000-2003) and the subsequent four years (2005-2008) was used to estimate these values. The race variables were split into White, Black, and Hispanic. These values summed to between 98% and 114% for each row. Adjustments to these three variables were made in the R-Code so that each row would sum to 100%. Because the first three values in this summary were often greater than or equal to 100%, Asian was omitted from this analysis. Subsequent analysis could include this variable. See Census Bureau definitions for further clarification.

Note, for all independent demographic/geographic variables used in this analysis, variables within each category sum to 100%, as noted above. For statistical modeling, this introduces multicollinearity (perfect correlation). As a result, adjustments were made in the R Code to account for this multicollinearity phenomenon. Several approaches were available, including removing one of the variables from the analysis for the variable group of interest. The other category is implicitly defined by the total summing to 100%. This approach was used for ANOVA and Multiple Regression Modeling. Several other transformations could be used, including ratios, log-transformed values, or center-scale variables; however, these methods were not used in this paper.

SECTION 2: Data – Describe (D-I-K-W: Information)

Overview: Combined Quintiles

First, in the description phase, simple linear regression was performed by combining all quintiles into one estimate and presenting the regression line's results in Table 1a below. From the table, note that the initial year of evaluation is 2000, and the y-intercept numbers represent year=0, so the y-intercept may not be a useful or readily interpretable value. Health insurance as a percentage of all costs increases on average across all income ranges is statistically significant ($p < 0.001$) at 0.165% per year, and in absolute 2023, \$USD amounts by \$128.92 per year, which is also statistically significant ($p < 0.001$). The explanatory power of the increase in health insurance percentage costs (Exp_Pct_Hlth_Ins) is considerably more reliable with an R^2 value of 0.6197 compared with the R^2 value of 0.3724 for absolute health insurance costs (Exp_USD_Hlth_Ins_2023_Adj).

Table 1a: Regression for all Quintiles Combined				
Dependent Variable	Year Coefficient (Slope: m)	Intercept (b)	p-value	Adjusted R^2
Exp_Pct_Hlth_Ins	0.16515	-327.6	< 0.001	0.6197
Exp_USD_Hlth_Ins_2023_Adj	128.92	-256158.28	< 0.001	0.3724

Note: * indicates statistical significance level:

* $p < 0.001$, $p < 0.01$, $p < 0.05$

Table 1a: Regression- Quintiles Combined 1

A quick visualization to get a sense of direction for the outcome variables, health insurance expenses (as a percentage of all household expenses) (Figure 1a), and health insurance costs (adjusted for 2023 \$USD) (Figure 1b) were produced and appear below. All quintiles were combined into one chart to represent the aggregated change of all quintile ranges (income strata) over the nearly twenty-five-year period of available data for health insurance expenses as a percentage of total household expenses. The weighted average method (each income quintile

range or stratum representing twenty percent of the total) was used to calculate the percentages, and the total dollar amounts are in the figures below. (SOM, 2024)

From the graphs below, increases are noted for health insurance cost measures, both as a percentage of all household expenses and as a raw dollar amount. The increases are noted in the upward slope of the actual values and the estimated regression line in the graphs. The following section, Summary Statistics, discusses the value increases and changes. During the COVID-19 pandemic from 2020-2023, one notable trend is the decrease in healthcare insurance expenses, both as a percentage of total household expenses and as an amount in the two figures below. Subsequent analysis will consider this observation from the perspective of income quintile groups. Was this the same for all income quintile groups, or did some income ranges vary regarding the impact seen from 2020-2023 in the figures below?

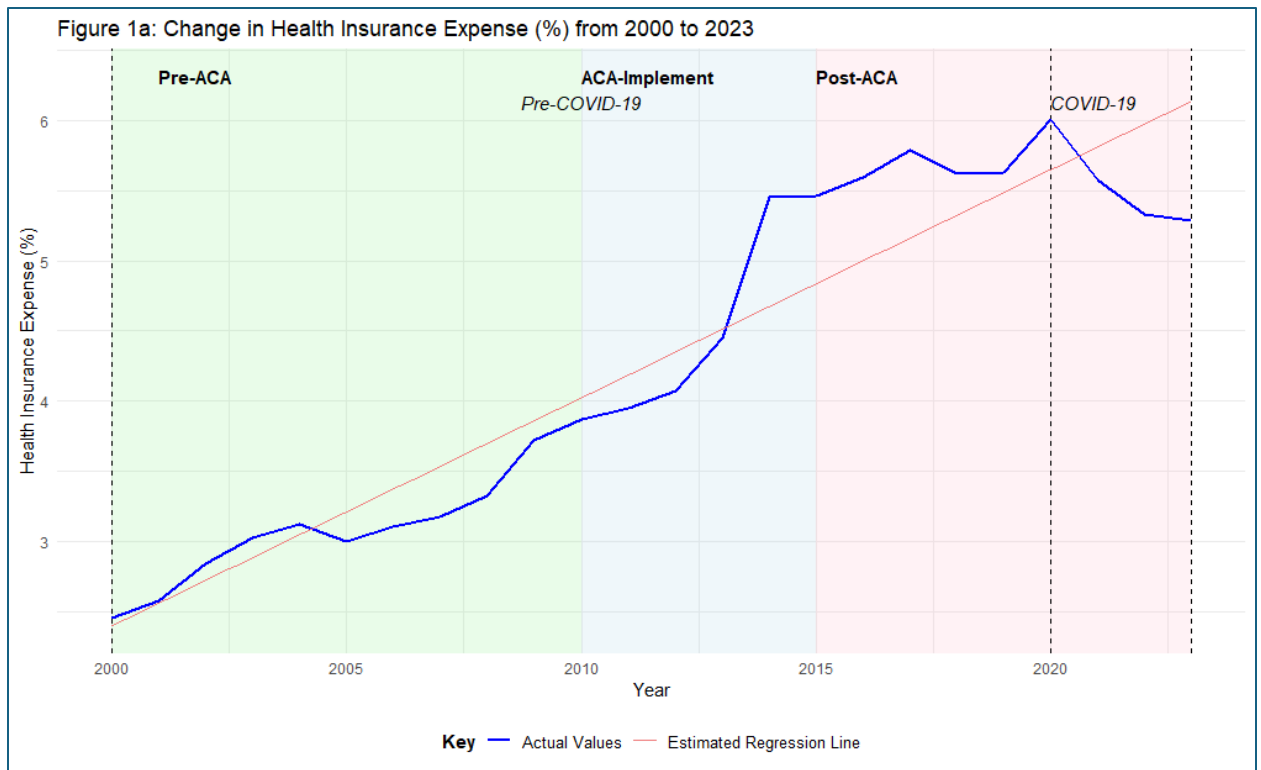


Figure 1a: Comb Inc Qnt Hlth Ins %

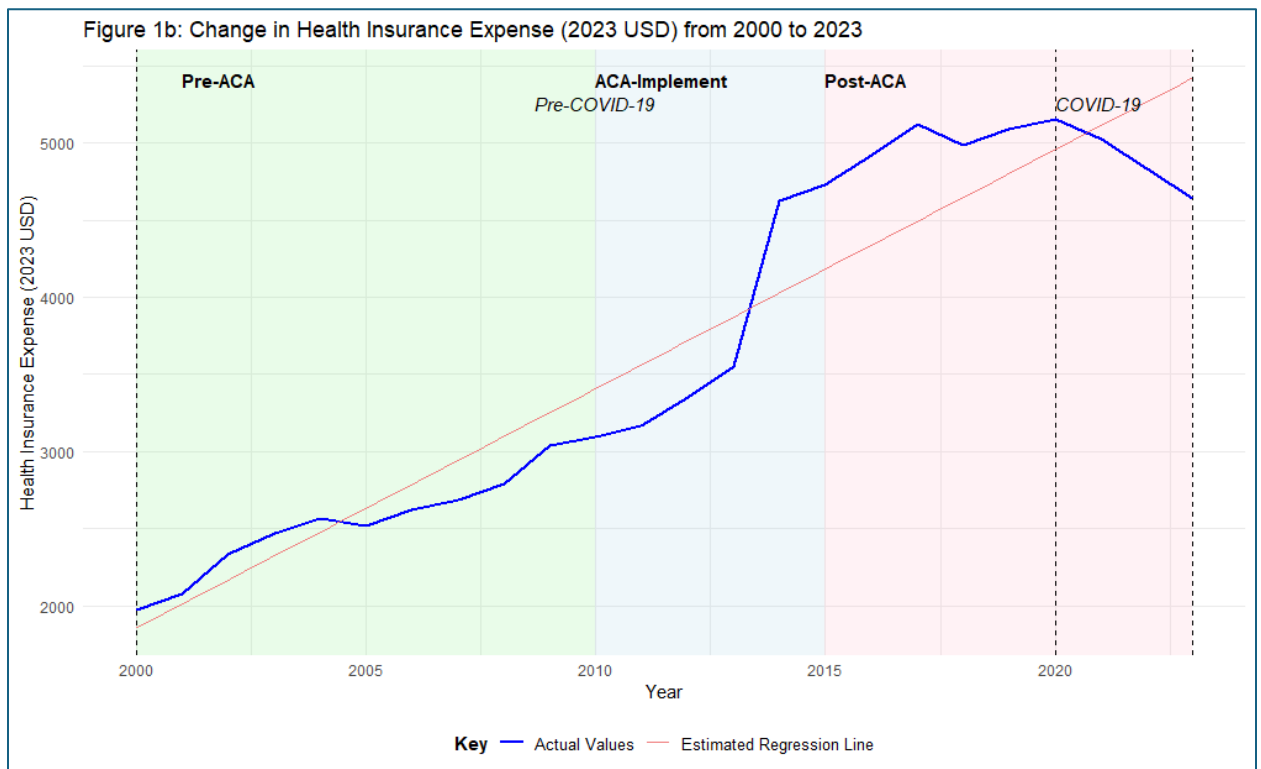


Fig 1b Comb Inc Qnt Hlth Ins \$

Summary Statistics

Summary statistics were generated for the y dependent variables, Exp_Pct_Hlth_Ins (Health Insurance as a Percentage of All Household Expenses) and Exp_USD_Hlth_Ins_2023_Adj (Health Insurance Costs in Real Values, Adjusted to 2023 values).

Referring to the first two tables below, Table 2a1 and Table 2a2, we see that Quintile 1 has the greatest maximum value with a range of 3.6% to 7.1%, and Quintile 5 has the lowest maximum value with a range of 1.7% to 4.8%. Health insurance is a higher percentage of total costs and has a wider range for those in the lowest income category compared to upper-income groups. For example, policies in the form of ACA subsidies may need to be adjusted for the lower income quintile ranges (Qnt1 and Qnt2) to alleviate the disproportionality.

Tables 2b1 and 2b2 show that the maximum expense in Qnt5 (\$6,971) is nearly 3 times greater than the maximum for the lowest Qnt1 (\$2,536). This is another indicator of spending capacity disparity. Overall, the percentage of total household expenditures on health insurance has risen disproportionately for lower-income households over the most recent twenty-three-year period. However, in absolute terms, costs have risen more for those in the upper-income ranges. This underscores the disproportionate burden placed on lower-income households.

Table 2a1: Overall Statistics for Exp_Pct_Hlth_Ins (in %)					
Variable	Mean	Median	Std_Dev	Min	Max
Exp_Pct_Hlth_Ins	4.59	4.50	1.45	1.70	7.40

Table 2a2: Overall Statistics for Exp_Pct_Hlth_Ins by Quintile (in %)					
Quintile	Mean (%)	Median (%)	Std Dev (%)	Min (%)	Max (%)
Qnt1	5.11	4.5	1.27	3.6	7.1
Qnt2	5.47	5.1	1.24	3.6	7.4
Qnt3	4.82	4.6	1.29	2.7	6.9
Qnt4	4.31	4.15	1.34	2.3	5.9
Qnt5	3.25	3.1	1.1	1.7	4.8

Table 2a1/2a2-Hlth Ins %- Summary Stats

Table 2b1: Overall Statistics for Exp_USD_Hlth_Ins_2023_Adj (in USD \$)					
Variable	Mean	Median	Std_Dev	Min	Max
Exp_USD_Hlth_Ins_2023_Adj	3,165.35	2,750.00	1,458.15	1,222.00	6,971.00

Table 2b2: Overall Statistics for Exp_USD_Hlth_Ins_2023_Adj by Quintile (in \$USD)					
Quintile	Mean (USD)	Median (USD)	Std Dev (USD)	Min (USD)	Max (USD)
Qnt1	1695	1472	490	1222	2536
Qnt2	2631	2305	665	1748	3600
Qnt3	3066	2754	840	1745	4353
Qnt4	3723	3442	1166	2017	5217
Qnt5	4712	4206	1694	2320	6971

Table 2b1/2b2: Hlth Ins \$ Summary Stats

Check Dependent Variables for Normal Distribution

Next, a check for the dependent variables approximation to be normally distributed before conducting correlation analysis was performed with a histogram and then QQ-Plots. See Figures 2a1 and 2b1- histograms of the dependent variables, and Figures 2a2 and 2b2 for the QQ-Plots. Both variables (Exp_Pct_Hlth_Ins and Exp_USD_Hlth_Ins_2023_Adj) are approximately normally distributed. However, from Figure 2b1: Health Insurance Cost in 2023 \$USD, a multi-modal distribution is noted with a primary peak near \$2000 and secondary peaks near \$4000 and \$6000. From the QQ-Plots, the points largely follow the red dashed line, which represents the theoretical normal distribution, indicating that the data is approximately normally distributed in the central range.

Figure 2a1: Distribution of Health Insurance as % of All Household Costs

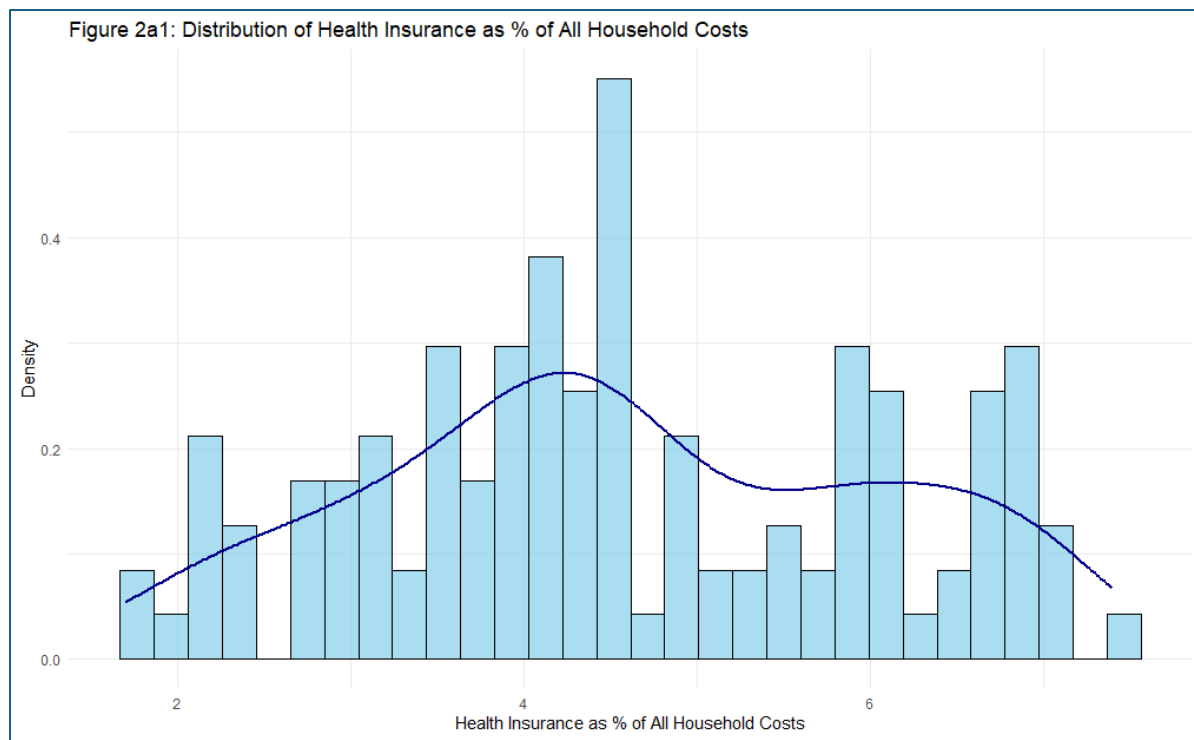


Figure 2a1: Distrib of Health Ins %

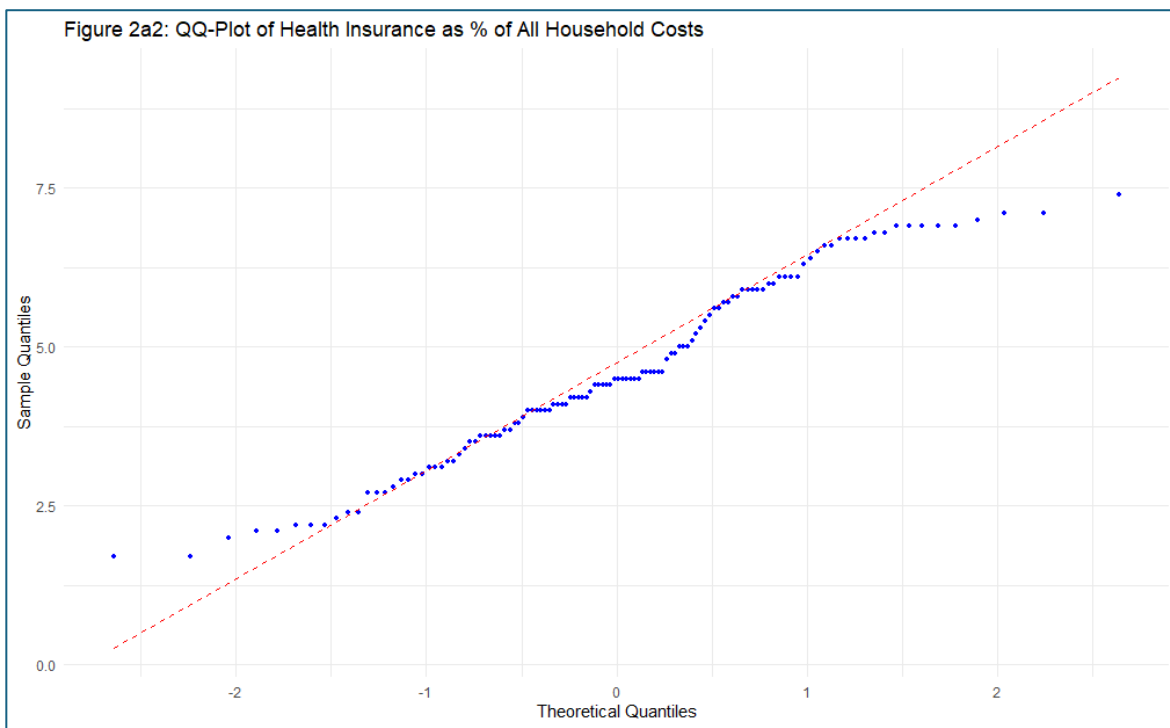


Figure 2a2: QQ-Plot of Health Ins %

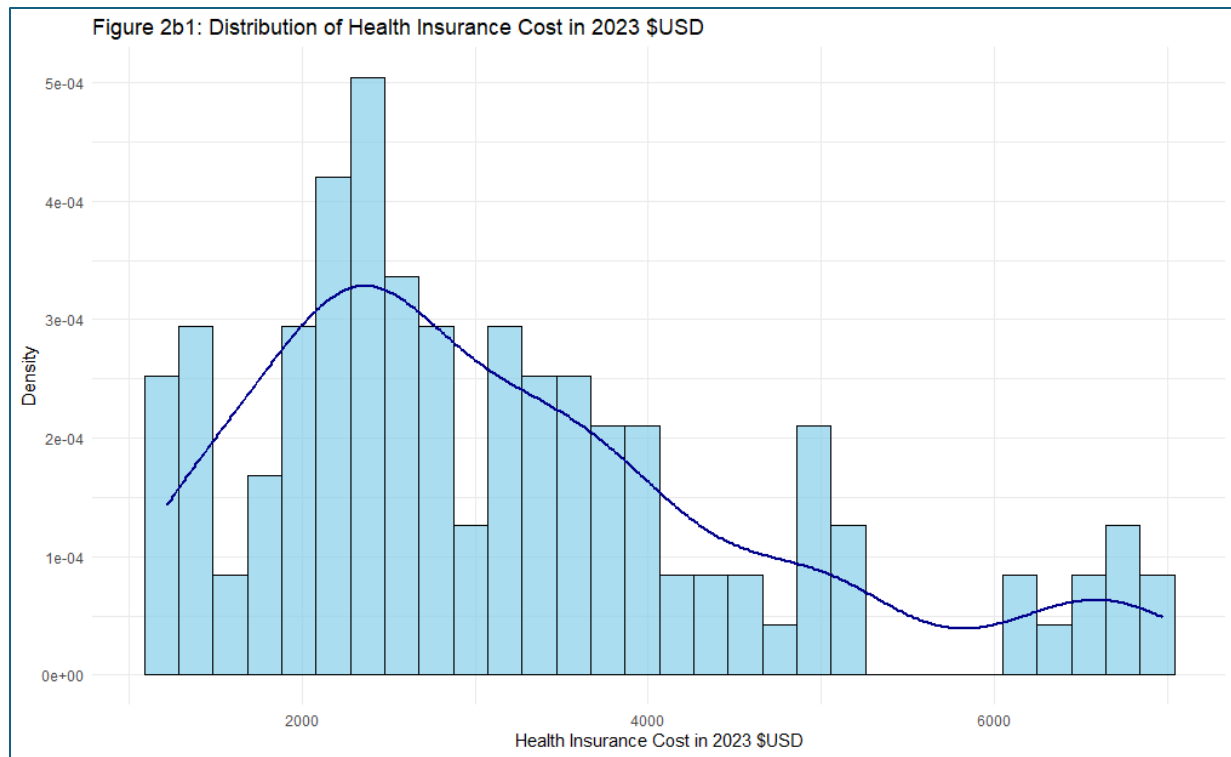


Figure 2b1: Distribution of Health Ins \$

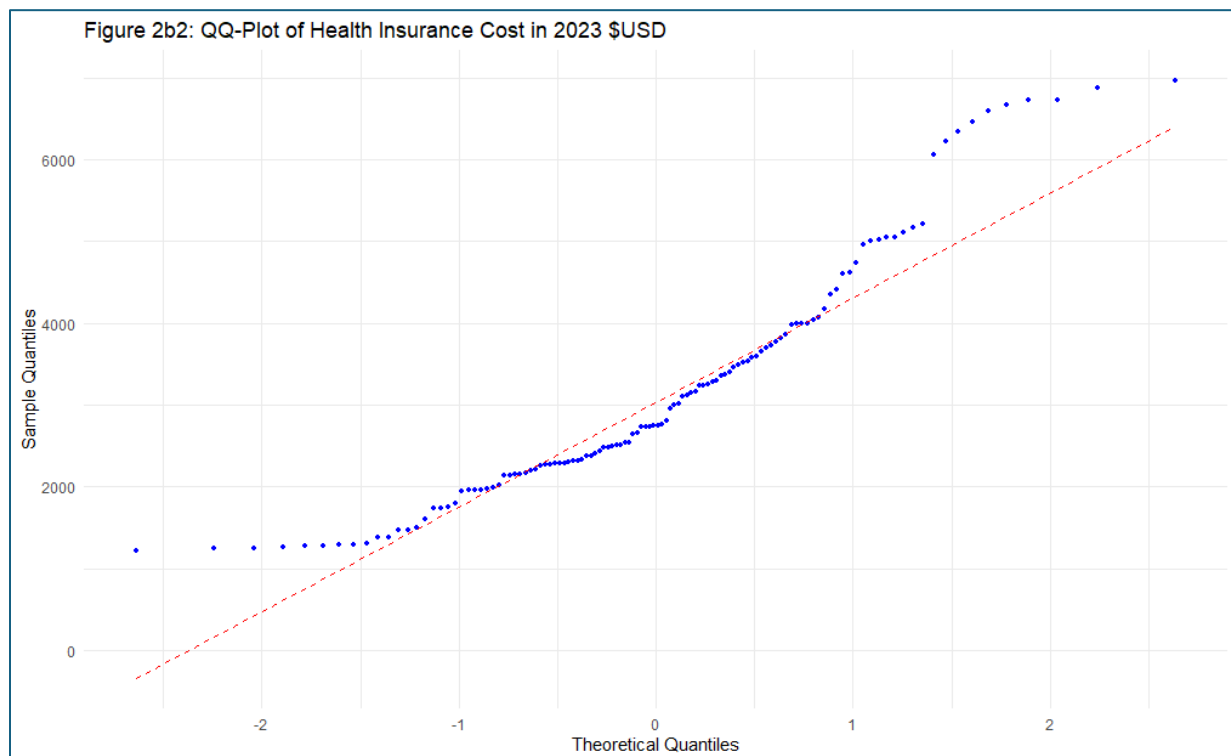


Figure 2b2: QQ-Plot Health Ins Costs \$

Correlation

Before beginning the correlation analysis, note that the demographic and regional variables, X1-X7, each consist of subcategories that sum to 100%. The subcategory summation to 100 % introduces the phenomena of multicollinearity. Because of multicollinearity, neither the Pearson nor Spearman Correlation Tests are ideal. However, to gain some insight into the relationships that may exist between the dependent variables and the dependent variables, the Spearman Correlation (which does not depend on a linear relationship among variables) was run on the dataset. All variables were included within each demographic and regional group for correlation analysis. Each demographic variable group (X1-X7) was compared against the dependent variables Exp_Pct_Hlth_Ins and Exp_USD_Hlth_Ins_2023_Adj. The results are represented in Figure 3 below.

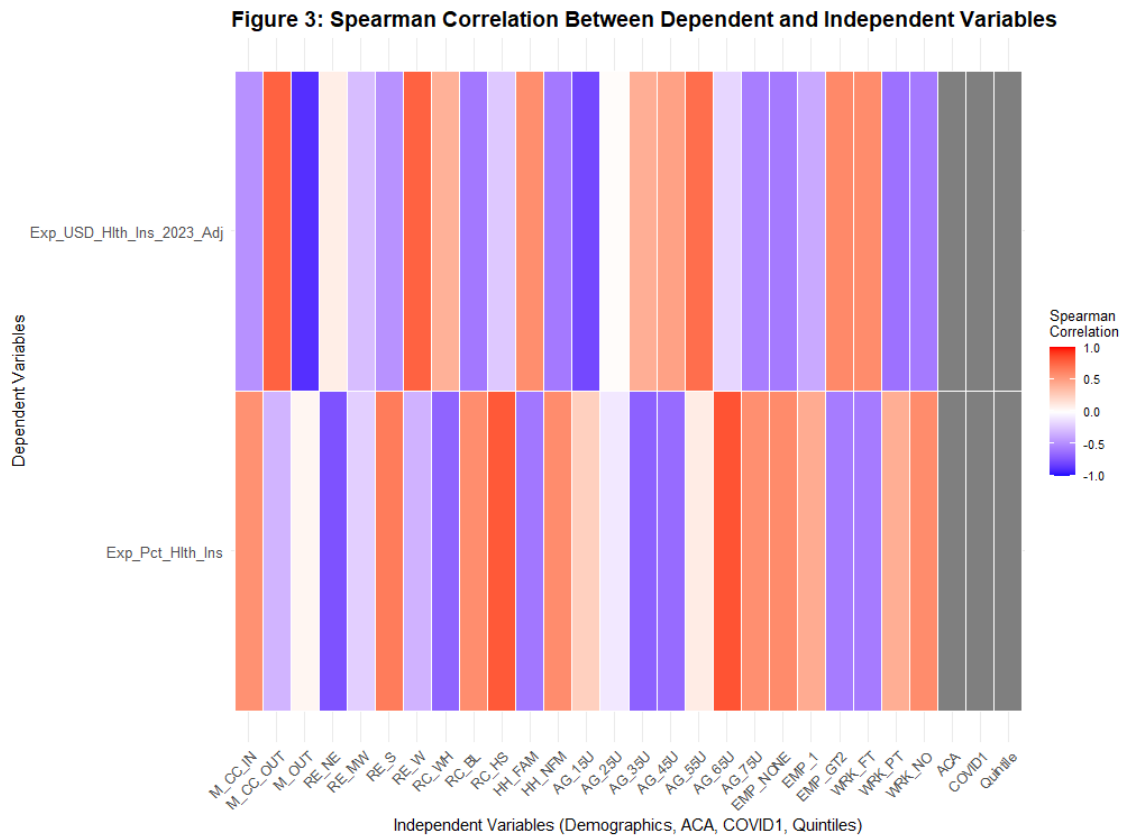


Figure 3: Spearman Correlation

Spearman Correlation

The Figure 3 Spearman Correlation color grid can be interpreted as follows:

Red implies strong positive correlations (close to +1), blue is associated with strong negative correlations (close to -1), and white indicates weak or no correlation (close to 0).

Some possible insights gained by quintile ranges include:

For the lower income quintiles, Qnt1 and Qnt2, there is a higher correlation between actual health insurance costs and households with no earners, no work type, and older age groups. A higher correlation indicates that lower-income households experience more significant financial hardship in obtaining or having health insurance. Households with more than two earners have a weaker correlation with lower absolute health insurance costs.

For the higher income quintiles, Qnt4 and Qnt5, more significant positive correlations exist between actual health insurance costs and the number of earners greater than two or working full-time. These positive correlations indicate higher absolute health insurance costs for wealthier households. Negative correlations between health insurance costs as a percent of all household costs and households with greater than two earners indicate a lower health insurance cost burden among households with higher incomes.

Box Plots

Next, boxplots were created for each dependent health insurance variable plotted first against ACA and Quintile, then against COVID-19 and Quintile. From the box plots (**Figure 3a and Figure 3b**) in the lower two income quintiles, health insurance represents a more significant portion of overall household costs throughout the ACA periods; however, the upper two income quintiles pay higher total health insurance costs across all ACA periods. During the COVID pandemic, health insurance as a percentage of all costs was higher for the lower two income

quintiles compared to the other quintiles; the upper-income quintiles paid more significant actual costs among all periods.

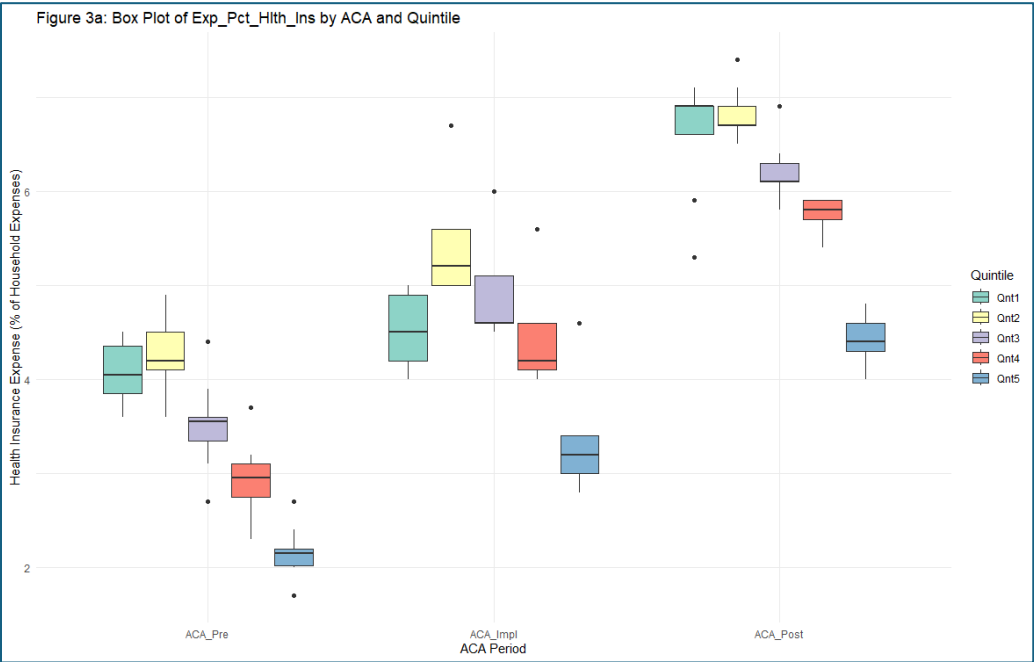


Fig 3a Box Plot Hlth Ins % ACA by Qnt

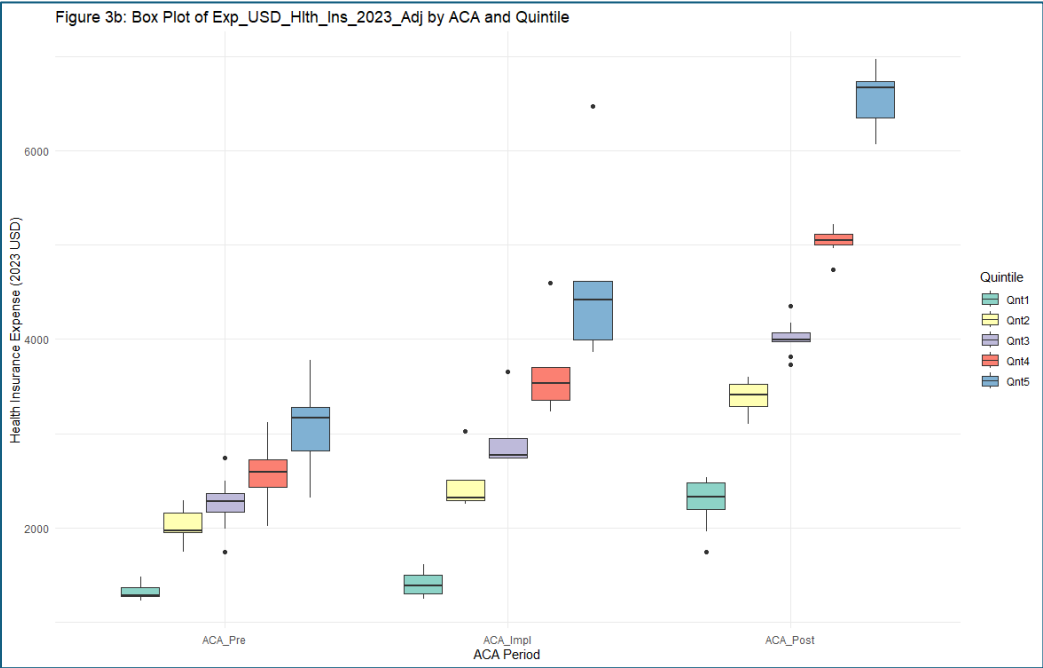


Fig 3b Box Ploth Hlth Ins \$ ACA by Qnt

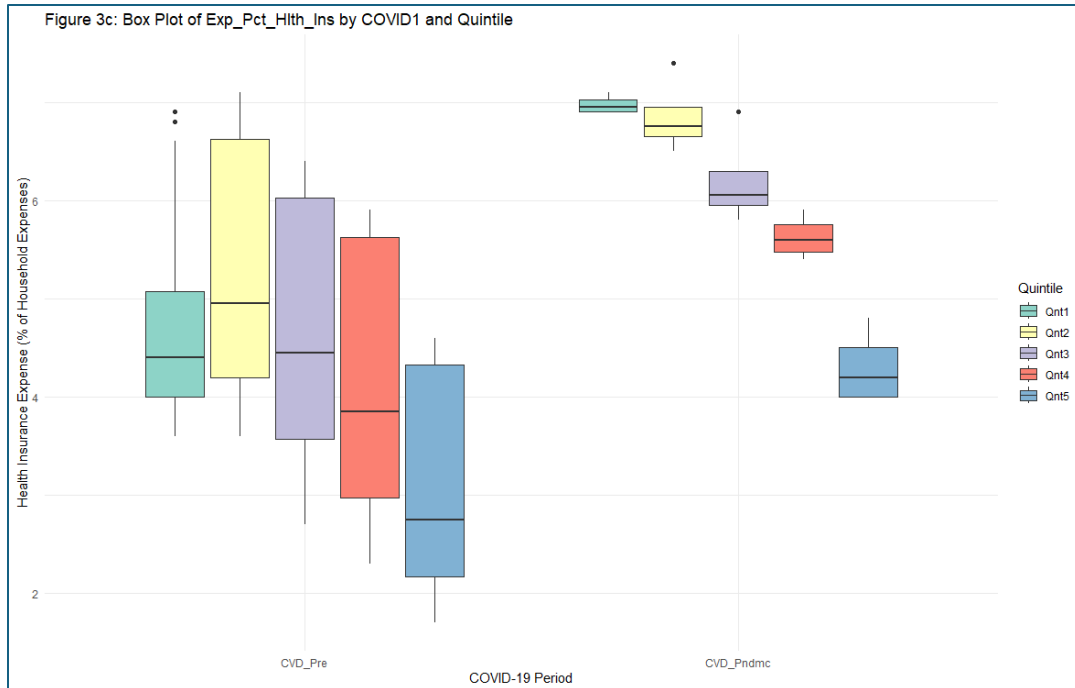


Fig 3c Box Plot Hlth Ins % COVID by Qnt

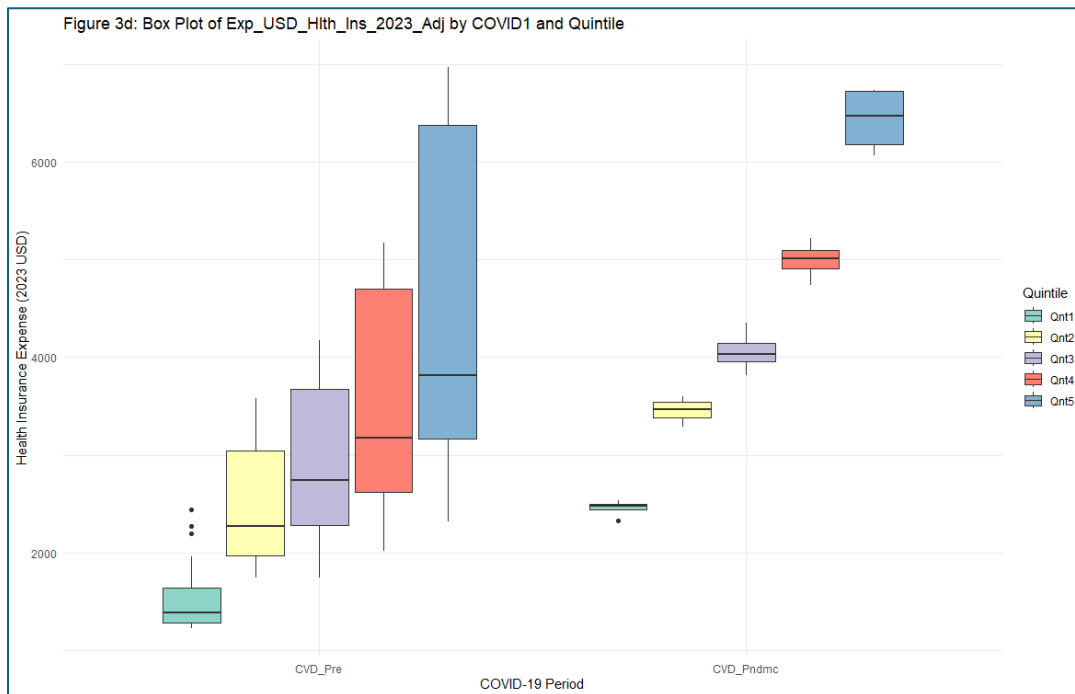


Fig 3d Box Plot Hlth Ins \$ COVID by Qnt

ANOVA

ANOVA analysis was limited due to the impact of multicollinearity on the stability and interpretability of the results. (Shrestha, 2020). ANOVA was performed with the dependent health insurance percentage and cost variables using the independent variables Year and Quintile. The results are shown in **Table 3b** below.

Hypothesis Tests

Hypothesis Tests and the results are stated below.

1. Exp_Pct_Hlth_Ins vs Year, Quintile

H_0 (Null Hypothesis): Year and Quintile have no effect on the % of household expenses spent on health insurance.

H_a (Alternative Hypothesis): Year and/or Quintile significantly affect the % of household expenses spent on health insurance.

2. Exp_USD_Hlth_Ins_2023_Adj vs Year, Quintile

H_0 (Null Hypothesis): Year and Quintile have no effect on health insurance costs adjusted to 2023 USD.

H_a (Alternative Hypothesis): Year and/or Quintile significantly affect health insurance costs adjusted to 2023 USD

The results of the hypothesis testing for all of the above four scenarios are: the null hypotheses (H_0) are rejected, with the predictors (Year and Quintile) significantly influencing the dependent health insurance variables. The Quintile variables consistently showed a higher impact than the year variable, underscoring the impact of income level on health insurance cost variations. However, the residuals indicate that the models could be improved by including more variables in the analysis. Data modeling will be undertaken in the next section of this project.

Table 3b: Summary of ANOVA Analysis							
Analysis	Factor	Df	Sum Sq	Mean Sq	F value	Pr(>F)	Significant?
1. Exp_Pct_Hlth_Ins ~ Year + Quintile	Year	23	172.45	7.498	90.86	<2e-16 **	Yes
	Quintile	4	71.71	17.927	217.24	<2e-16 **	Yes
	Residuals	92	7.59	0.083			
2. Exp_USD_Hlth_Ins_2023_Adj ~ Year + Quintile	Year	23	105,807,009	4,600,305	18.1	<2e-16 **	Yes
	Quintile	4	123,832,884	30,958,221	121.8	<2e-16 **	Yes
	Residuals	92	23,378,567	254,115			
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1							

Table 3b: Summary of ANOVA Year/Quintile

SECTION 3: Data – Model (D-I-K-W: Knowledge)

Simple Linear Regression

To begin the modeling phase, a simple linear regression graph of our two dependent variables of interest, health insurance as a percentage of total household expenses (Exp_Pct_Hlth_Ins) and health insurance actual cost in 2023 adjusted US dollars (Exp_USD_Hlth_Ins_2023_Adj), will be graphed by year and quintile range. The ACA and COVID-19 periods are highlighted first in the regression equations in Tables 4a and 4b and then illustrated in Figures 4a and 4b below. From the tables, note that the initial year of evaluation is 2020, and the y-intercept numbers represent year=0, so the y-intercept may not be a useful-readily-interpretable value.

The slopes are of interest, noting that for the health insurance percentage variables, the slopes for the lowest four income quintiles vary by increasing on average 16.5% to 17% per year, and most notable is the highest income quintile five, showing an average 14% increase across all years and this was the lowest percent increase of all income ranges. Note that all quintiles were statistically significant, with a p-value <0.001 for each. The Adjusted R² was 0.62 for Quintile 1, 0.87 for Quintiles 2-4, and 0.84 for Quintile 5. To interpret the meaning of the “slope values,” consider Table 2a2, where the lowest quintile had a minimum value of 3.6% in 2000 and a

maximum value of 7.1% in 2020. The average increase over these twenty years is approximately 16.7% per year.

For actual health insurance costs (2023 adjusted), note the increases represented by the slopes (representing the average increase in dollar amounts per year over the twenty-three-year period) in the tables are lowest for quintile one at \$62.56 per year and gradually increase to \$223.72 per year for quintile five, the uppermost quintile. The explanatory power or R^2 value is considerably higher for the model when the income levels are split by quintile ranges versus the R^2 values from the combined quintile ranges. The Adjusted R^2 values range from 0.8067 to .8996. They are more significant for the regression lines or health insurance percentage variable estimates. The R^2 values increase by two to nearly two-and-a-half times from 0.3776 to 0.8996 and are significant (p -value<0.001) across all quintiles for the actual health insurance costs (adjusted to 2023 \$USD).

Table 4a: Regression Equations for Health Insurance %						
Quintile	Year Coefficient (Slope: m)	Intercept (b)	Std. Error (Year)	t-value	p-value	Adjusted R²
Qnt1	0.16515	-331.6	0.01183	13.96	<0.001	0.6197
Qnt2	0.16561	-327.65	0.01256	13.19	<0.001	0.8761
Qnt3	0.17139	-331.3	0.0134	12.79	<0.001	0.8761
Qnt4	0.17139	-353.3	0.0134	12.79	<0.001	0.8761
Qnt5	0.1437	-285.8	0.01288	11.16	<0.001	0.8431

Note: * indicates statistical significance level:

*p < 0.001, p < 0.01, p < 0.05

Table 4a: Regression Health Ins % by Qnt

Table 4b: Regression Equations for Health Insurance in 2023 \$						
Quintile	Year Coefficient (Slope: m)	Intercept (b)	Std. Error (Year)	t-value	p-value	Adjusted R²
Qnt1	62.56	-124140.78	6.3528	9.847	<0.001	0.8067
Qnt2	88.84	-176065.33	6.5779	13.506	<0.001	0.8875
Qnt3	112.98	-224202.27	7.8506	14.392	<0.001	0.8996
Qnt4	156.5	-311081.41	11.1104	14.086	<0.001	0.8957
Qnt5	223.72	-445301.61	18.2272	12.274	<0.001	0.8668

Note: * indicates statistical significance level:

*p < 0.001, p < 0.01, p < 0.05

Table 4b Regression Health Ins \$ by Qnt

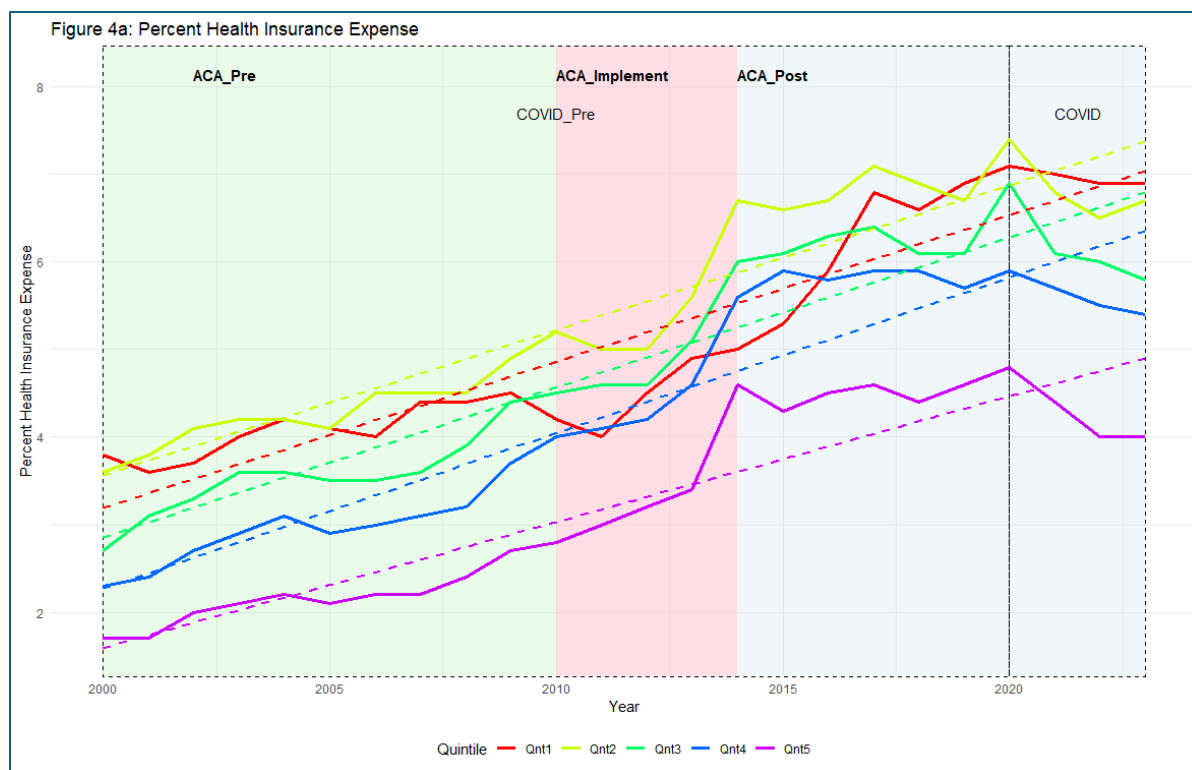


Figure 4a Health Ins % by Year/Quintile

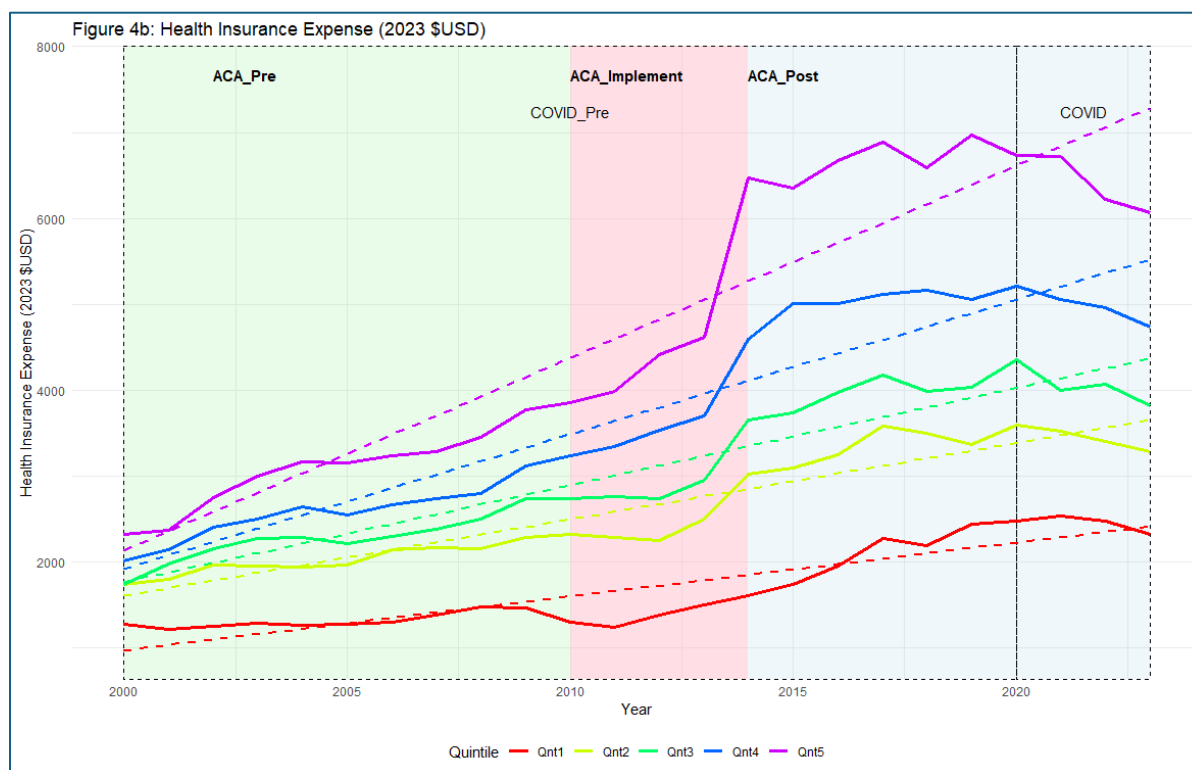


Figure 4b Health Ins \$ by Year/Quintile

Multiple Linear Regression

Next, multiple linear regression was attempted as a modeling approach. However, due to multicollinearity with the year variable and the temporal variables (ACA and COVID1) and multicollinearity in each subcategory comprising the demographic and regional variables under consideration for this project, multiple regression was ruled out as a viable modeling option. Sequential Variable Removal (Indicator Variable Deletion) (Kutner, Nachtsheim, Neter, Li, 2005) and Variance Inflation Factor (VIF) Analysis were attempted to correct or adjust for the multicollinearity phenomena.

First, Sequential Variable Removal (Indicator Variable Deletion) was attempted, where one of the subcategories for each independent variable (X1-X7) was removed. However, a significant drop in the R2 value was noted, indicating the relevance of the deleted variables. For example, M_OUT (Non-metropolitan) in X1 and EMP_NONE in X6 had a negative impact on the R2 value. This method had minimal impact on reducing multicollinearity among the remaining variables.

Next, the VIF method was calculated for all independent variables (X1–X7), yielding a VIF factor > 5. A VIF factor > 5 indicates problematic multicollinearity. This method also had minimal impact on reducing multicollinearity among the variables used in the model, resulting in the need to consider other modeling options.

Machine Learning (ML)

The project consists of a multitude of explanatory or independent variables, many of which express multicollinear relationships. For example, the temporal variables ACA and COVID-19 are based on the Year variable. Also, each of the regional and demographic variables, X1-X7 (metropolitan area, regional area, race, household type, age, employment (number of earners), and work type)) consist of subcategories within each of these groups that sum to 100%. For this reason, Machine Learning, which has several algorithms or processes minimally affected by multicollinearity, was chosen as the modeling approach. The analysis will begin with Random Forest, and then, for final variable selection, Ridge Regression will be employed.

ML: Random-Forest / Ridge Regression, All Quintiles, Combined

First, Health Insurance Percent and then Health Insurance (adj 2023 \$USD) for all 26 demographic independent variables was evaluated using the Machine Learning technique, Random Forest. (*NOTE: Due to changing definitions over the 2000-2023 time period for definitions of Race variable categories (notably including a Hispanic mix with other subcategories and adding the Asian subcategory), the Race variable was excluded from the Machine Learning modeling exercise to enhance the clarity of the end result variable selection process or decision.) Note that the training dataset comprises 70%, and the test dataset is 30% of the rows in the dfACA01 “R-data frame.”

The variable selection process employed by combining the Random Forest and Ridge Regression techniques is visually presented below. To begin with, see the variable comparison results in Figure 5a below. In Figures 5a - Health Insurance as % of all Household Costs and Figure 7a -Health Insurance Cost (adj 2023 \$USD), the %IncMSE (Percent Increase in Mean

Square Error) and IncNodePurity(Increase in Node Purity) measures were computed and ranked or sorted for each of the 26 independent variables.

Next, the Random Forest iterative selection technique was employed, where fifty iterations of computations of the measures “%IncMSE” and “IncNodePurity” were taken, aggregated, and averaged. Then, the ranked top ten independent variables were selected. The top ten variables for each dependent variable are represented in Figures 5b, Health Insurance Percent, and 7b, Health Insurance Cost. Common descriptive variables among both dependent variables include metro locations (M_OUT, non-metro or M_IN, in metro), age variables (AG_65U – age 65 and older, up to 74; and AG_35 – age 35 and up to 44), and regional locations (Western and North East Regions). Two variables, age 45 up to 54 and Northeast Region, are found as explanatory variables for Health Insurance %, but not Health Insurance Cost. Likewise, M_CC_IN (Inside City Center) and RE_W (Western region) are found to be indicators for Health Insurance Costs but do not appear in Health Insurance %.

Finally, after the first two steps of the Random Forest methodology, Ridge Regression was incorporated into the descriptive variable reduction exercise to determine the top five variables that could be used for explanatory purposes of each of the dependent Health Insurance variables.

Regarding model performance, the explanatory power of the five variables selected for Health Insurance % resulted in an R^2 (the coefficient of determination) value of 0.9136, compared to an R^2 value of 0.7979 for the Health Insurance Cost variable. The R^2 value measures the proportion or amount of variation measured in the dependent variables that the independent variables in the model can explain. The Mean Absolute Error (MAE) (average magnitude of prediction errors) and the Root Mean Square Error (RMSE) (square root of the

average squared prediction error – more significant errors have greater weight and are used to emphasize the effect of outliers) (Sabbha, 2024) are also shown in **Table 5c** and **Table 7c**. These values are intuitively smaller for Health Insurance % because of the scale of the values for this variable. The more significant errors shown in Table 7c for Health Insurance Cost reflect the more considerable variability in this dependent variable. Because RMSE is greater than MAE in both models, this indicates the presence of outliers but is more significant for Health Insurance Costs.

Among the top five independent variables selected as the most significant predictors in the models, these four variables appear in both as top 5 indicators: Non-metro (M_OUT), Metro area (M_IN), age 65 to 74 (65_U), age 35 to 44 (AG_35U). The Northeast region (RE_NE) is exclusive to the Health Insurance % variable, whereas outside of the city center (M_CC_OUT) is unique as a contributor to explaining absolute costs, highlighting geographic differences in metro areas. **Table 9** below lists the variables that are the dominant drivers of Health Insurance %'s and Costs, underscoring the role of age and geographic or metro location in determining Health Insurance Cost levels.

However, it is important to note some limitations of Ridge Regression. Ridge Regression is primarily used for variable selection and addressing multicollinearity and involves selecting an appropriate tuning parameter (λ). This parameter significantly influences variable selection and the model's final performance; thus, choosing an optimal λ can be subjective and require cross-validation. Additionally, Ridge Regression produces biased estimates, as the coefficients are intentionally reduced towards zero to decrease variance and improve stability, potentially complicating the interpretation of individual variable impacts. These limitations must

be considered when interpreting Ridge Regression results, primarily when used for explanatory modeling purposes. (Zhu, 2024)

Table 9: ML: Demographic Variable Comparison for Health Insurance % and Cost		
Feature	Health Ins % Model	Health Ins Cost Model
Dominant Driver	Age variables (e.g., AG_65U, AG_35U)	Metro and geographic variables (e.g., M_OUT, M_IN)
Role of Age	Critical for proportional spending, reflecting demographic-driven burdens.	Secondary, influencing absolute costs but less so than geographic factors.
Role of Geography	Regional differences matter for proportions (e.g., RE_NE).	Metro and non-metro locations dominate, reflecting structural pricing.

Table 9 Demg Var Compare Hlth Ins % vs \$

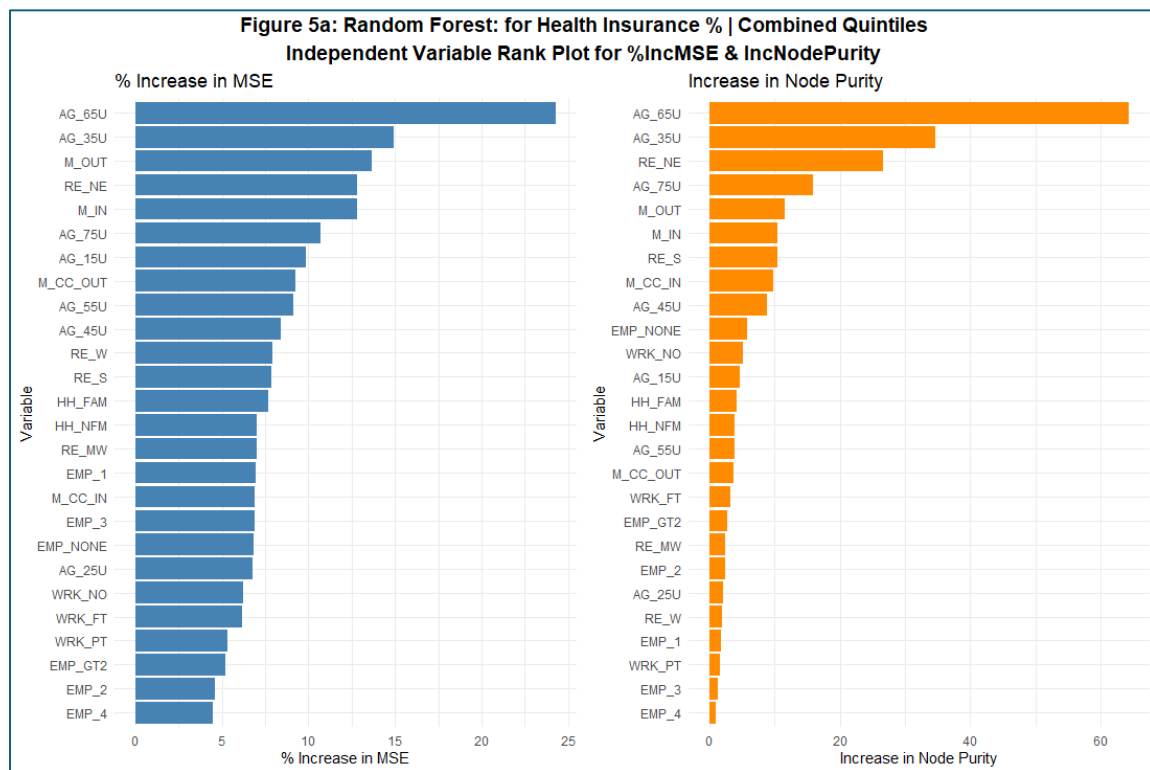


Fig 5a HLRF Hlth Ins % Comb Qnt All Vars

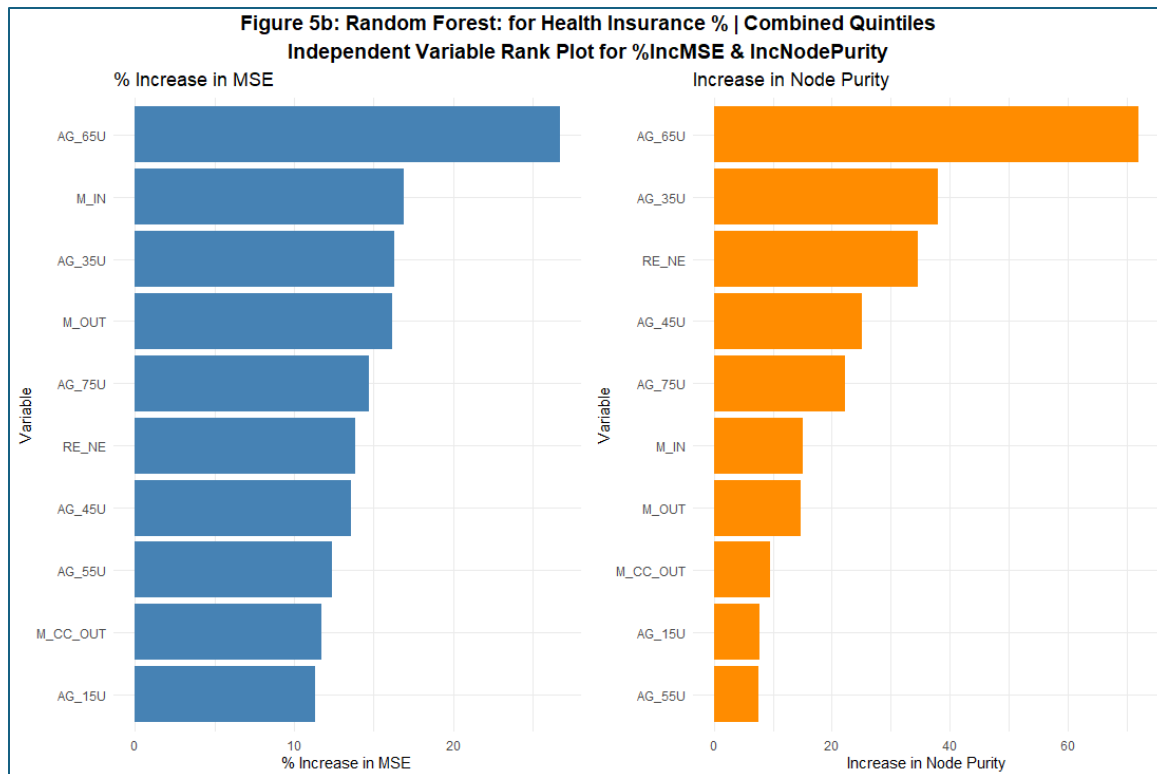


Fig 5b MLRF Hlth Ins % Comb Qnt Top 10

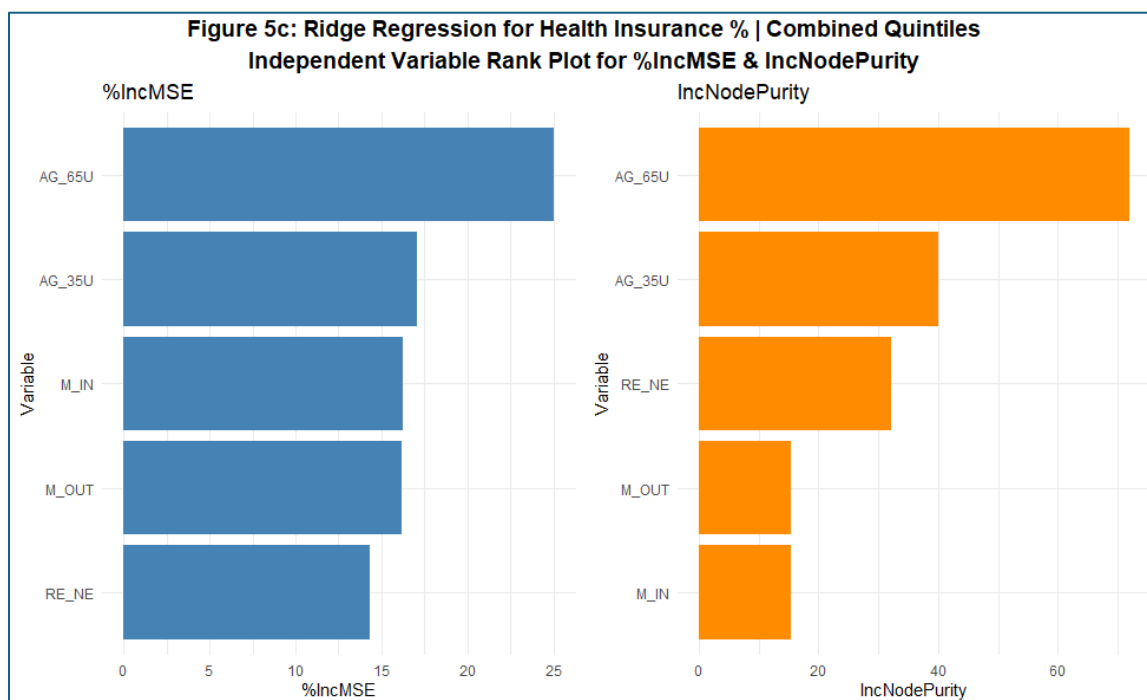


Fig 5c Ridge Hlth Inx % Comb Qnt Top 5

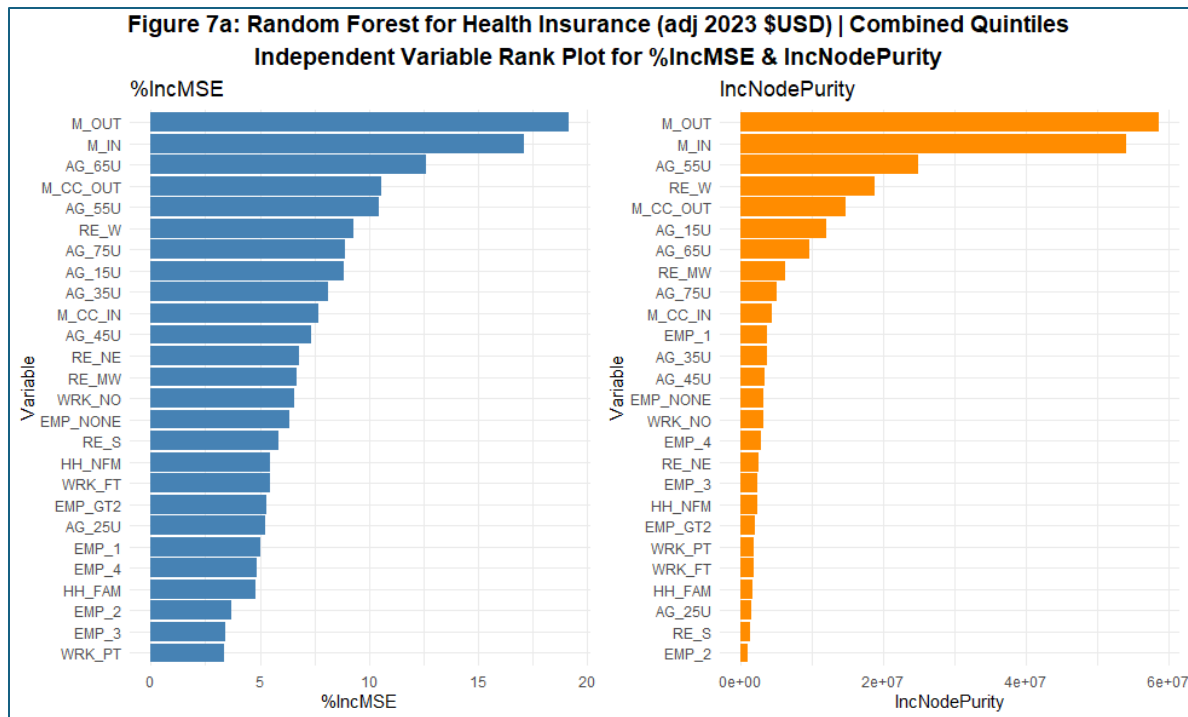


Fig 7a MLRF Hlth Ins \$ Comb Qnt All Vars

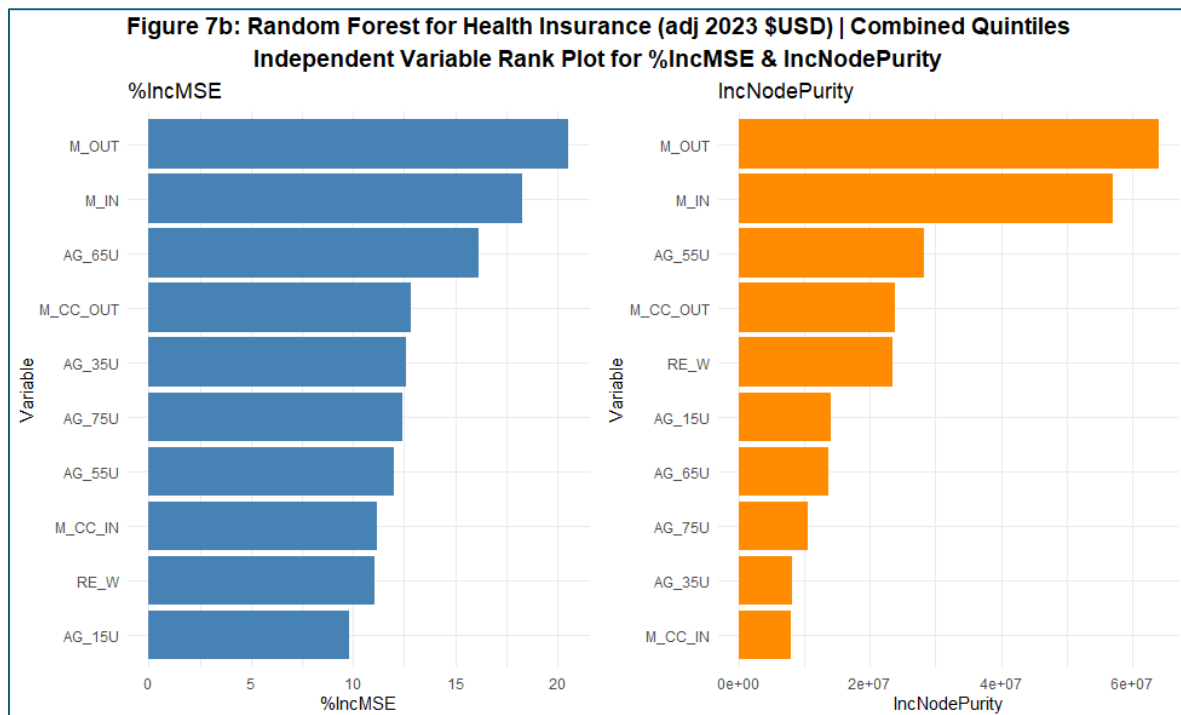


Fig 7b MLRF Hlth Ins \$ Comb Qnt Top 10

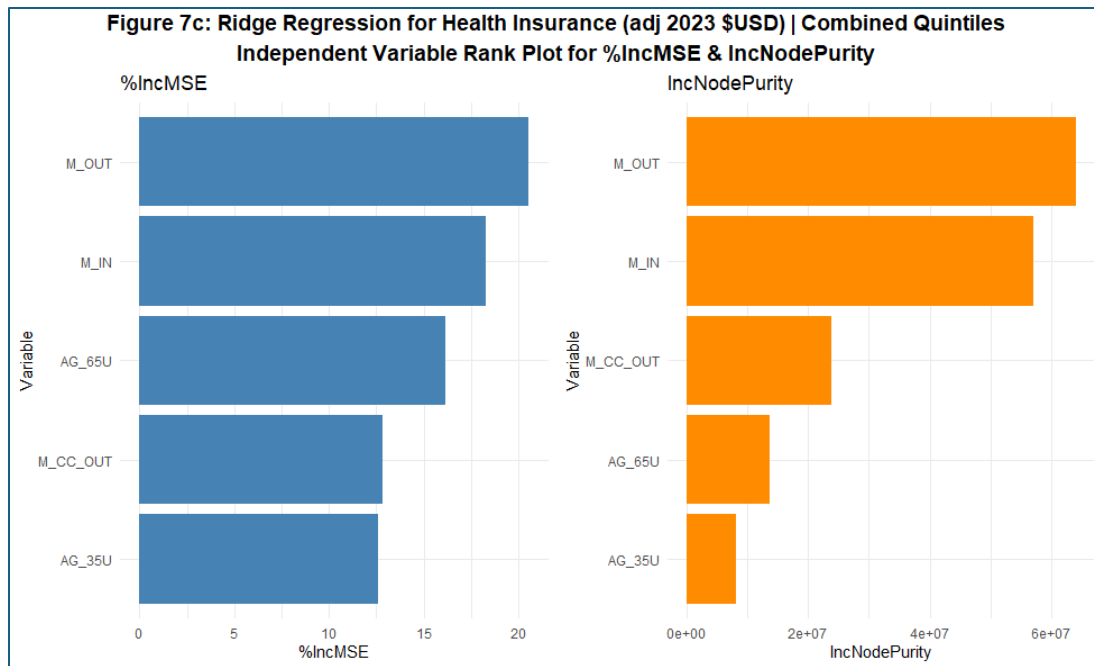


Fig 7c Ridge Hlth Ins \$ Comb Qnt Top 5

SCENARIO 1: Health Ins % : Combined Quintiles

Table 5a

Run 1: Random Forest Health Ins % All 26 Indep Vars	
Metric	Value
R-squared (R^2)	0.9423992
Mean Absolute Error (MAE)	0.2735054
Root Mean Squared Error (RMSE)	0.3476263
Number of Variables	26

Table 5b

Run2: Random Forest Health Ins % Top 10 Indep Vars	
Metric	Value
Average R-squared (R^2)	0.9466524
Average Mean Absolute Error (MAE)	0.2573329
Average Root Mean Squared Error (RMSE)	0.3345246
Number of Variables	10
Number of Iterations	50

Table 5c

Run3: Ridge Regression Health Ins % Top 5 Indep Vars			
Metric	Value		
R-squared (R^2)	0.91359074		
Mean Absolute Error (MAE)	0.335925288		
Root Mean Squared Error (RMSE)	0.425773163		
Number of Variables	5		
	Final 5 Vars:	%IncMSE:	IncNodePurity
	AG 65U	25.06	71.9
	AG 35U	17.09	40.11
	M OUT	16.2	15.49
	RE NE	14.29	32.15
	M IN	16.25	15.38

Tbl 5a-c RF Ridge Hlth Ins % Metrics 1

SCENARIO 3: Health Ins (adj 2023 \$USD) : Combined Quintiles

Table 7a

Run 1: Random Forest Health Ins (adj 2023 \$USD) All 26 Indep Vars	
Metric	Value
R-squared (R^2)	0.9487483
Mean Absolute Error (MAE)	216.639877
Root Mean Squared Error (RMSE)	328.7302703
Number of Variables	26

Table 7b

Run2: Random Forest Health Ins (adj 2023 \$USD) Top 10 Indep Vars	
Metric	Value
Average R-squared (R^2)	0.9502422
Average Mean Absolute Error (MAE)	200.8128291
Average Root Mean Squared Error (RMSE)	323.8503277
Number of Variables	10
Number of Iterations	50

Table 7c

Run3: Ridge Regression Health Ins (adj 2023 \$USD) Top 5 Indep Vars			
Metric	Value		
R-squared (R²)	0.797928022		
Mean Absolute Error (MAE)	473.5162996		
Root Mean Squared Error (RMSE)	652.737445		
Number of Variables	5		
	Final 5 Vars:	%IncMSE :	IncNodePurity:
	M OUT	20.56	64061179.36
	M IN	18.3	57065111.93
	AG 65U	16.17	13661376.17
	M CC OUT	12.82	23881789.35
	AG 35U	12.62	8178362.06

Tbl 7a-c RF Ridge Hlth Ins \$ Metrics

ML: Random-Forest / Ridge Regression, by Quintile

Next, the same machine learning algorithm approach (Random Forest to reduce the number of demographic and regional variables from 22 to 10 and then Ridge Regression (to more effectively ignore multicollinearity) to narrow down the explanatory variables to the top 5) was followed as in the previous section. This time, the quintile range analysis was run separately instead of combined for all quintiles.

The results reflecting R^2 values for the models by quintile are shown in Table 8A1 (Table 8A1 Rigdge Top 5 Indep Vars for Health Ins % (Exp_Pct_Hlth_Ins)) and Table 8A2 (Table 8B1 Rigdge Top 5 Health Ins \$ (Exp_USD_Hlth_Ins_2023_Adj)), below. Tables 8B1 and 8B2 also list the top five independent variables by quintile for Health Insurance Percentages and Cost Amounts, showing the variable metrics of “%IncMSE” and “IncNodePurity.” Finally, Table 8C (Health Insurance % Vs. \$ Variable Comparison) lists the differences in indicator variables for the quintile analysis for Health Insurance Percentage and Total Costs. The following is a discussion of comparisons within the Health Ins % and Cost \$ Tables, and the results are shown in Table 8C.

Regarding Health Insurance as a Percentage of All Costs, the age 65-74 category (AG_65U) is the common top 5 variable across all income quintile ranges. For Health Insurance Actual Cost, the age 45-54 (AG_45U) category is the common variable across all income ranges. This finding indicates that for each Health Insurance cost measure, the upper age category is a consistent driver of the percentage cost increase. In contrast, the middle age category is a consistent driver in total health insurance cost. No common variables exist across all income quintiles for the health insurance cost measures; however, age and regional differences are the common drivers of cost increases in both health insurance cost metrics.

Table 8A1 Rigdge Top 5 Indep Vars for Health Ins % (Exp_Pct_Hlth_Ins)				
Quintile	R²	RMSE	MAE	Top Variables
Qnt1	0.96	0.24	0.17	M_OUT', 'AG_35U', 'M_CC_OUT', 'AG_65U', 'AG_15U'
Qnt2	0.95	0.27	0.24	M_CC_OUT', 'AG_55U', 'RE_MW', 'AG_65U', 'HH_FAM'
Qnt3	0.91	0.38	0.31	AG_35U', 'WRK_NO', 'AG_65U', 'RE_NE', 'AG_75U'
Qnt4	0.95	0.29	0.24	AG_45U', 'AG_65U', 'WRK_NO', 'RE_W', 'M_OUT'
Qnt5	0.95	0.25	0.21	AG_65U', 'AG_35U', 'AG_45U', 'WRK_NO', 'M_CC_IN'

Tbl 8a1 Hlth Ins % Top 5 by Qnt

Table 8B1 Rigdge Top 5 Health Ins \$ (Exp USD Hlth Ins 2023 Adj)				
Quintile	R²	RMSE	MAE	Top Variables
Qnt1	0.97	91.78	73.11	AG_35U', 'AG_25U', 'RE_NE', 'AG_15U', 'AG_45U'
Qnt2	0.95	163.59	132.62	RE_MW', 'AG_15U', 'M_CC_OUT', 'AG_45U', 'AG_65U'
Qnt3	0.89	282.94	222.89	RE_NE', 'AG_75U', 'M_CC_OUT', 'AG_45U', 'M_OUT'
Qnt4	0.95	264.07	211.00	RE_W', 'AG_75U', 'M_OUT', 'AG_45U', 'EMP_NONE'
Qnt5	0.95	384.84	332.90	EMP_NONE', 'AG_65U', 'M_OUT', 'AG_45U', 'AG_35U'

Tbl 8b1 Hlth Ins \$ Ridge Top 5 by Qnt

Table 8a2 Ridge Top 5 Health Ins %			
Quintile	Variable	%IncMSE	IncNodePurity
Qnt1	M_OUT	7.919067	3.485549
	AG_35U	6.87592	2.878311
	M_CC_OUT	8.400107	3.254467
	AG_65U	8.465104	4.789927
	AG_15U	9.389315	6.716332
Qnt2	M_CC_OUT	8.462818	4.2256973
	AG_55U	8.936258	0.7447622
	RE_MW	5.74788	2.7876537
	AG_65U	8.188606	3.8730345
	HH_FAM	10.439598	6.1331612
Qnt3	AG_35U	8.311353	3.596785
	WRK_NO	7.384187	1.747341
	AG_65U	11.846256	7.773972
	RE_NE	4.682331	1.362591
	AG_75U	10.995059	6.933334
Qnt4	AG_45U	8.688367	4.900979
	AG_65U	11.765123	7.943462
	WRK_NO	9.066673	4.913541
	RE_W	5.913364	1.121623
	M_OUT	4.986622	1.084149
Qnt5	AG_65U	10.681592	4.9130836
	AG_35U	6.897318	1.1310843
	AG_45U	10.981186	4.7974558
	WRK_NO	5.571144	0.9037203
	M_CC_IN	5.917462	1.3343892

Tbl 8a2 Ridge Hlth Ins % Top 5

Table 8b2 Ridge Top 5 Health Ins \$			
Quintile	Variable	%IncMSE	IncNodePurity
Qnt1	AG_35U	5.4355	250076.3
	AG_25U	6.163	328270.7
	RE_NE	4.8561	154132.5
	AG_15U	7.9784	564134.4
	AG_45U	5.1455	258683.6
Qnt2	RE_MW	2.6938	260384
	AG_15U	5.4276	117820.7
	M_CC_OUT	12.5178	2437517
	AG_45U	5.3429	522003.2
	AG_65U	10.7709	1904792
Qnt3	RE_NE	5.9531	865426.9
	AG_75U	11.103	2643107
	M_CC_OUT	8.853	2023457
	AG_45U	8.4682	2119719
	M_OUT	7.5754	1441543
Qnt4	RE_W	7.9939	2393664
	AG_75U	9.2007	4087809
	M_OUT	6.2095	1799664
	AG_45U	11.4896	5774384
	EMP_NONE	9.9988	4690417
Qnt5	EMP_NONE	8.5173	9698143
	AG_65U	12.9183	15602556
	M_OUT	6.4784	5127048
	AG_45U	9.922	9845305
	AG_35U	5.3069	1753595

Tbl 8b2 Ridge Hlth Ins \$ Top 5

Table 8C: Health Insurance % Vs \$ Variable Comparison					
Quintile	Health Ins % Var	Health Ins \$ Var	Vars In Both Health % & \$	Vars Unique to Health Ins %	Vars Unique to Health Ins \$
Qnt1	M_OUT, AG_35U, M_CC_OUT, AG_65U, AG_15U	AG_35U, AG_25U, RE_NE, AG_15U, AG_45U	AG_35U, AG_15U	M_OUT, M_CC_OUT	AG_25U, RE_NE, AG_45U
Qnt2	M_CC_OUT, AG_55U, RE_MW, AG_65U, HH_FAM	RE_MW, AG_15U, M_CC_OUT, AG_45U, AG_65U	RE_MW, AG_65U, M_CC_OUT	AG_55U, HH_FAM	AG_15U, AG_45U
Qnt3	AG_35U, WRK_NO, AG_65U, RE_NE, AG_75U	RE_NE, AG_75U, M_CC_OUT, AG_45U, M_OUT	RE_NE, AG_75U	AG_35U, WRK_NO	M_CC_OUT, AG_45U, M_OUT
Qnt4	AG_45U, AG_65U, WRK_NO, RE_W, M_OUT	RE_W, AG_75U, M_OUT, AG_45U, EMP_NONE	RE_W, M_OUT, AG_45U	AG_65U, WRK_NO	AG_75U, EMP_NONE
Qnt5	AG_65U, AG_35U, AG_45U, WRK_NO, M_CC_IN	EMP_NONE, AG_65U, M_OUT, AG_45U, AG_35U	AG_65U, AG_45U, AG_35U	WRK_NO, M_CC_IN	EMP_NONE, M_OUT

Tbl 8c Hlth Ins % vs \$ Ridge Top 5

ANCOVA: Evaluating the Impact Temporal Variables, ACA and COVID1

For the final analysis, ANCOVA testing was conducted for the dependent variables Health Insurance %'s (Exp_Pct_Hlth_Ins) and Health Insurance Costs (Exp_USD_Hlth_Ins_2023_Adj) and analyzing effects of the independent variables: ACA, COVID1, and Quintile. Code runs were labeled as follows: Health Insurance % as ANCOVA1, with Scenario1 grouping all quintiles and Scenario2 listing results for each quintile (income group) separately. The analysis for health insurance cost was referred to as ANCOVA2, and the same two scenarios were run as ANCOVA1. The results of the ANCOVA testing are presented below in Tables 10a and 10b. Following is a discussion of significant results.

First are the metric definitions (stated again for clarity):

Metric Definitions:

- ACA: Evaluates the effect of the Affordable Care Act periods (Pre, Implementation, Post).
- COVID1: Assesses the impact of pandemic phases (Pre-pandemic, Pandemic).
- Quintile: Analyzes differences across income quintiles (Qnt1–Qnt5).
- Residual Mean Sq: Represents unexplained variance after accounting for the predictors.

A summary of findings for the combined quintiles (Table 10a) follows. ACA was significant in health insurance variable analysis (% and cost). A stronger effect was found for health insurance % (proportional terms - ANCOVA1: $F=384.036$) than in dollar terms (ANCOVA2: $F=169.265$). COVID-19 analysis yielded no significant findings for either health insurance-dependent variable. The Quintile variable was significant in both analyses, indicating that income levels strongly influence health insurance as a percentage of all household costs and dollar value amounts. The health insurance percentage

variable had a lower residual variance, implying that the model better fits health insurance as a percentage versus health insurance in actual dollar amounts.

The key findings of **Table 10b** indicate that ACA is significant across all quintiles for both health insurance percentage and cost (proportional and dollar terms). The F-values are more significant in the dollar-level analysis across income quintiles, which reflects the greater scale of the financial impact relative to income level. The COVID1 variable (COVID-19) p-values were determined to be significant for the lowest income quintile, QNT1, for both health insurance variables. The p-value variance indicates more limited pandemic effects on higher-income groups. The residual variance was lower for the health insurance percentage variable, indicating a better model fit. The residuals increased in value as quintiles increased for the health insurance cost variable, implying higher variability in dollar terms.

In terms of the null hypothesis, we have the following. First, for the combined income quintiles analysis,

H₀ The null hypothesis is: The mean of the dependent health insurance variables is the same across all groups, ACA, COVID-19(Covid-1 variable), and income quintiles.

H_a: The alternative hypothesis is: The mean of the dependent health insurance variables differ across all groups, ACA, COVID-19(Covid-1 variable), and income quintiles.

ANCOVA Hypothesis Findings:

H_0 is rejected for ACA and Quintiles for Health Insurance Metrics % and Cost.

The rejection of H_0 indicates that ACA and Quintiles significantly impact the Health Insurance cost metrics.

H_0 is not rejected for COVID-19 for the Health Insurance Metrics: % and Cost.

This indicates that there is no evidence that COVID-19 had a significant impact on Health Insurance cost metrics.

Next, for hypothesis testing for the temporal/categorical variables of ACA and COVID-19, which were evaluated by income quintile, we have the following:

General Hypothesis for ACA and COVID-19 by Quintile Evaluation:

H_0 : Null Hypothesis: ACA and/or COVID-19 does not significantly impact the dependent variable health insurance cost metrics under evaluation.

H_a : Alternative Hypothesis: ACA and/or COVID-19 significantly impacts the dependent variable health insurance cost metrics under evaluation.

Hypothesis testing results:

ACA: H_0 is rejected for the Health Insurance metrics across all income quintiles.

The interpretation is that the ACA period significantly impacts health insurance metrics across all income quintiles.

COVID-19: H_0 is rejected for both Health Insurance metrics for income quintile

Qnt1. The interpretation is that the COVID-19 period significantly impacts both health insurance metrics across all income quintiles.

H_0 is not rejected for income quintiles Qnt2-5, indicating that

COVID-19 has no significant impact on the health insurance metrics.

Table 10a: ANCOVA Results for Health Ins % and Cost, Quintiles Combined							
Metric	ACA (F-value)	ACA (p-value)	COVID1 (F-value)	COVID1 (p-value)	Quintile (F-value)	Quintile (p-value)	Residual Mean Sq
ANCOVA1 (Health Ins %)	384.036	<2e-16 ***	0.237	0.628	87.651	<2e-16 ***	0.2
ANCOVA2 (Health Ins Cost)	169.265	<2e-16 ***	0.044	0.833	107.977	<2e-16 ***	286713

Tbl 10a ANCOVA Hlth Ins % & \$ Comb Qnt

Table 10b: ANCOVA Results for Health Ins % and Cost, Quintiles Separated							
Metric	Quintile	ACA (F-value)	ACA (p-value)	COVID1 (F-value)	COVID1 (p-value)		Residual Mean Sq
ANCOVA1 (Health Ins %)	Qnt1	93.057	7.4e-11 ***	5.792	0.0259 *		0.175
	Qnt2	80.3	2.77e-10 ***	0.028	0.868		0.197
	Qnt3	81.51	2.43e-10 ***	0	1		0.209
	Qnt4	112.314	1.33e-11 ***	0.611	0.443		0.168
	Qnt5	74.218	5.57e-10 ***	0.435	0.517		0.165
ANCOVA2 (Health Ins Cost)	Qnt1	97.89	4.68e-11 ***	10.08	0.00476 **		24448
	Qnt2	106.228	2.22e-11 ***	0.456	0.507		43667
	Qnt3	96.058	5.55e-11 ***	0.172	0.683		76498
	Qnt4	130.202	3.41e-12 ***	0.123	0.729		111539
	Qnt5	85.225	1.63e-10 ***	0.424	0.522		345588

Tbl 10b ANCOV Hlth Ins % & \$ by Qnt

SECTION 4: Data – Findings (D-I-K-W: Wisdom)

Key findings from the analysis in this report are divided into three sections, or Findings.

Finding 1: Trends over time for health insurance costs as a percentage of all household costs and pure health insurance costs in absolute dollar values have increased over the twenty-three-year period from 2000 to 2023. The cost increases over time are represented graphically in Figures 1a and 1b (metrics with aggregated income ranges) and 4a and 4b (metrics represented by the five income quintiles). Health insurance costs as a percentage of all household costs have increased more for the lower-income households (Quintiles 1 and 2) versus the upper-income households (Quintile 4 to some extent and, more significantly, Quintile 5). Health insurance costs in pure dollar value amounts (adjusted for 2023 \$ values) increased significantly for the upper-income Quintile 5, especially compared with income Quintiles 1 and 2.

Finding 2: Demographic impacts measured by the machine learning techniques Random Forest and Ridge Regression indicate that age and regional location are the most influential demographic variables. In the combined quintile analysis, age groups 65 to 74 and 35 to 44 were the most significant predictors of Health Insurance Percentage Costs. These two age groups were also in the top 5 Health Care Dollar Cost predictors but were less influential for this metric. Metro "In" and "Out" were the highest predictors of Health Insurance costs. These two indicators were among the top five indicators in the model for health insurance percentages but had a lower impact. The North East region was found to be in the top five influential variables for Health Insurance %'s but did not appear in the top five variables for actual health insurance costs. Likewise, In-Metro, but not in the city center (M_CC_OUT), was in the top five indicators for actual health insurance costs but did not appear in the health insurance percentage measure.

A per-quintile analysis of the demographic and regional variable impact on health insurance costs was also conducted using machine learning with random forest and ridge regression techniques. These approaches were utilized to narrow the field of twenty-two possible cost predictors to the top five cost predictors for each income quintile. This analysis also yielded age (in this case, the categories of age 65-74 and age 45-54) and regional location to be significant influences or have the greatest impact on health insurance costs over the twenty-three-year period of this study. Household structure, employment type, and status were among the top five influences varying by income quintile range.

Finding 3: The ACA "appears" to have significantly influenced US citizens overall from a health insurance cost perspective and across all five income groups. COVID-19 significantly impacted US Citizens in the lowest income group (Qnt1) compared to all other income groups (Qnt2-5). Further analysis may need to be run to verify this statement, given the inter-relationship of the ACA and COVID-19(COVID-19) variables with the Year variable used in this study. Further ANCOVA analysis could be conducted to include the regional and demographic variables found in the "Findings 2" section above. Other statistical techniques, including Machine Learning or others, may need to be considered to determine the impact of ACA and COVID-19 on Health Insurance Costs in the US.

Discussion

Year and Income Quintile Influence:

Health insurance costs have increased from 2000 to 2023 on average across all income quintiles by approximately \$129 per year, based on the fit of standard linear regression (Table 1a). From a per-quintile perspective, health insurance has increased, varying from approximately \$63 per year for individuals in the lowest income quintile (Qnt1) compared to \$224 per year for

individuals in the upper-income quintile (Qnt5). However, for health care insurance as a percentage of all household costs, individuals in the lowest income quintiles have seen the most significant percentage increase compared to individuals in the upper-income levels. This is noted in Figure 4a graphically, and numerically in Table 4a. Also, the increases in health insurance costs as a percentage of all household costs are shown in the summary statistics in Table 2a2. Here, the lowest income quintile saw an increase for this metric, going from 3.6% to 7.1% over the period of interest, while those in the upper-income level noted a low of 1.7% to 4.8% across the same period. The highest income quintile also had the lowest health insurance cost-to-all-cost ratio of 4.8%, the lowest value across all income levels.

Demographic and Regional Influences:

The two demographic and regional variables most closely related to increases in health insurance costs are age and regional location. Demographic and regional influences were in the top five indicators when all income quintiles were combined or analyzed separately using the Random Forest and Ridge Regression Machine Learning techniques. When all income levels were combined, application of the ML algorithms to the study data set indicated that the age categories of 65 to 74 (AG_65U) and 35 to 44 (AG_35U) and the regional variables metro inside and metro outside (M_IN and M_OUT) were the most impactful variables related to the health insurance cost metrics. When the income quintiles were evaluated separately, the age category 65 to 74 (AG_65U) was most impactful for all income quintiles for health insurance as a percentage of all costs. In contrast, the age category of 45 to 54 (AG_45U) was the most important relative to actual health insurance costs (adjusted for 2023 \$USD). Regional and metro locations were also highly relevant but varied in impact due to income quintile.

The two largest age cohorts in the USA in 2023, Millennials (Gen Y, who were in the 27- to 42-year age range) and Baby Boomers (59- —to 77-year-olds and 36- —to 54-year-olds in the year 2000, at the start of this study period), likely had a major impact on the age ranges identified in this document. (Beresford Research, 2025)

Also, the US has experienced a significant increase or shift toward urbanization over the last twenty-plus years (Fry, 2020). This trend is likely reflected in the metro and regional trends identified as health insurance cost indicators in this report.

Temporal Variable Impact

The ANCOVA statistical measure was employed to measure the impact of this report's temporal variables of interest (ACA and COVID-19). ANCOVA, including additional impactful variables (from the Machine Learning exercise in this project), may offer more insight. Other statistical techniques, including Machine Learning, may be better suited to answer this question and could be incorporated in future studies. As of the data available in this report (2023), the COVID-19 pandemic was ending. Several years, five to ten, for example, may need to transpire before an accurate assessment of the impact of COVID-19 on health insurance costs.

Bias- Study Specific

Although a variable indicating Race was initially included in the analysis for this report, this variable was later excluded from analysis (specifically from the Modeling/Machine Learning portion of the research) due to ambiguity or changes in definitions over time of this variable in the Census Bureau's Table HINC-05. For example, the Asian category was not initially available in 2000. Also, the definitions of an individual identifying with one or multiple races and the Hispanic relation may have changed over the 2000 to 2023 period. A wealth of race information is available in the USCB's Table HINC-05; however, additional time would need to be devoted

to determining how best to utilize one or more interpretations of this variable to incorporate into this research paper. From an ethical and transparent perspective, it is important to understand why this variable was omitted from this study. However, with further investigation and "data wrangling," the race variable can most likely and certainly be included in studies of demographic factors impacting health insurance costs.

Also, in the USCB's Table HINC-05, variables describing Earners, Work Experience, and Household Tenure and Structure are provided and could be utilized in demographic impact analysis for health insurance costs. These four variable types were omitted from this project to reduce the complexity of the modeling but could be incorporated in future studies.

Future Research

Several takeaways about the breadth of data covered in this project include the variables in Table 1- Variable Descriptions of this report. As identified in the previous section about transparency, additional variables such as Race, earners, work experience, household tenure, and structure from the USCB's Table HINC-05 could be included. However, it will require more time regarding data wrangling and analysis to include at this time. Other resources for variables such as education and more detailed regional or city, county, and state data may also be available. Depending upon the audience of interest and the research question(s) being addressed would determine which additional variables to include. Data availability is another crucial first step to consider when asking for or requesting research on a particular topic.

ANCOVA evaluation of the temporal variables (ACA and COVID-19) was undertaken for this project; however, more detailed (better) statistical methods might be available to answer this question. Perhaps incorporating one or several other machine learning techniques may assist in a more careful and accurate assessment of these temporal variables.

From an income quintile perspective, this paper focused on the impact of temporal, demographic, and regional variables related to US Health Insurance costs over 2000-2023. The ACA has specific guidelines for premium tax credits, Medicaid expansion, and cost-sharing reductions (such as out-of-pocket costs) relative to Federal Poverty Level (FPL) guidelines. (111th Congress (2009-2010)). To more accurately assess the impact of the ACA in relation to FPL guidelines, additional data (from the USCB or other agencies) may be available first to obtain and then include in the analysis income level demographic information on a decile or percentile level. Such income level details may be necessary to assess the ACA's impact more accurately, which lawmakers, the general public, news organizations, health insurers, and others may most certainly want to know.

Conclusion

Simple line regression proved the best method to visualize and quantify health insurance cost increases and changes over time by income levels. Machine Learning techniques (Random Forest and Ridge Regression) proved to be the best methods to evaluate the impact of demographic and regional determinants on health insurance costs. ANCOVA analysis provided some insight into the impact of the temporal variables (ACA and COVID-19) used in this study; however, due to the multicollinearity of these two variables with the Year variable, it is difficult to access the actual numeric impact of these independent variable categories on health insurance costs. Other statistical methods may be more appropriate or thorough to address this impact question.

Data Science techniques including data wrangling (finding data sources, accounting for missing data, joining data from different sources, and other techniques), data discovery (correlation, ANOVA), data visualization (numerous tables and graphs), and data modeling

(Regression, Machine Learning, and ANCOVA) procedures were incorporated to produce this Capstone Project Report. The full gamut of Data Science procedures covered in UTICA University's Master's in Data Science program were incorporated in great detail in producing and synthesizing the results and report for this study. The author is very thankful for the opportunity to revisit many of these concepts and learn new techniques to incorporate into his years of experience in the data field.

Hopefully, those reading this capstone will find the results presented (including the underlying painstaking detail applied to produce the results) are not only appreciated but may prove to be very useful in providing insights into one of the most important topics on many US Citizens minds—health care costs—and what is impacting them. The report offers valuable insight that lawmakers, US citizens (paying for or trying to obtain health insurance), news agencies, and health care insurance institutions may utilize in making important decisions.

For further information, questions, comments, concerns, data, or analysis requests, please contact the author at STAT_X1@Yahoo.com.

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